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Anaerobic Digestion of Dairy Dissolved Air Flotation Sludge: Assessment of Enhancement Strategies

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Abstract

Dissolved air flotation (DAF), a primary treatment for high-strength waste (HSW) dairy wastewater, generates sludge with a high organic content, posing disposal challenges. Anaerobic digestion (AD) offers a solution to promote renewable energy use and enhance the circular economy of the dairy industry by converting this sludge into biogas. However, DAF sludge has low digestibility, potential inhibitory risks, nutritional imbalances, and other challenges. This study identified potential technological solutions to optimize DAF sludge use by enhancing AD performance, including pretreatment, anaerobic co-digestion (ACoD), additives, and process modifications, which have been successfully implemented for other organic substrates. Using modified weighted scoring analysis (WSA), microaeration pretreatment was found to be a superior method, waste-activated sludge was identified as the most suitable co-substrate, and bentonite was selected as the most promising additive. Two-stage anaerobic digestion (TSAD) emerged as the most promising comprehensive solution for process modifications, addressing predominant DAF sludge AD issues and other subsidiary challenges.

Keywords: Anaerobic digestion, Biogas, Dairy industry, Dissolved air flotation, Weighted scoring analysis

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1 Introduction

The dairy industry is a key global sector, with production reaching 965.7 million tonnes in 2023, marking a 1.5 % increase from 2022, according to the Food and Agriculture Organization (FAO) [1]. The rapid growth of dairy products has led to significant environmental challenges, particularly in waste management, where addressing organic pollutants is critical for reducing their environmental impact. As the industry grows, so does the volume of wastewater, with 2.5–10 L produced for every 1 L of processed milk [2, 3]. Although these pollutants typically do not cause severe issues, mismanagement can lead to problems, making strict control essential, preventing system failures and environmental harm [4].

Dairy wastewater (DWW) is rich in organic matter, such as carbohydrates, proteins, and fats, as shown by its high chemical oxygen demand (COD) from 0.79 to 68.6 g L^{-1} [5]. It offers recycling potential, but its composition varies with the processing methods, technologies, and cleaning procedures [5]. This wastewater often contains residual cleaners such as nitric acid and sodium hydroxide [5, 6]. The use of acidic and alkaline cleaning agents impacts pH levels and turns nutrient-rich wastewater into an environmental problem, contributing to eutrophication, acidification, emissions, soil degradation, and health risks [7–9].

To effectively treat their high-strength waste (HSW), dairy industries utilize the dissolved air flotation (DAF) process to primarily separate solids, organic matter, fat, oil, grease (FOG), and proteins before discharge or directed to further subsequent treatment [10–13]. However, this process generates DAF sludge, also known as DAF waste, DAF float, DAF slurry, DAF solid, or flotation sludge (FS) [7, 12, 14, 15]. Land spreading is the most widely applied disposal method for DAF sludge after dewatering [13, 16]. However, due to its high COD, high solid content,

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high fat, ammonia, nutrient content, and low pH, disposal of this material presents environmental challenges, increasing disposal costs and often requiring energy-intensive treatments, especially during wet weather when land application as a fertilizer is not feasible or advisable [17].

Due to its valuable organic matter, DAF sludge has the potential for resource recovery through anaerobic digestion (AD), struvite production, biochar, and vermicomposting [18–24]. Among these, AD is a promising simultaneous energy recovery and waste reduction, making it a sustainable and efficient solution [5]. However, the characteristics of DAF sludge introduce some key challenges that must be addressed to achieve a stable AD process.

Aside from DWW, DAF sludge can also be produced from different primary sources and can vary considerably in composition, depending on whether it originates from dairy farms, dairy processing industries, or meat processing facilities. In dairy farms, DAF is occasionally applied after AD for nutrient recovery [25, 26], whereas in the dairy and meat processing industries, it is typically employed as a primary wastewater treatment step [20]. These variations affect the organic matter content, nutrient levels, particle size, and sludge biodegradability. Nonetheless, all DAF sludges share a common generation mechanism based on physicochemical flotation, which results in comparable challenges for AD.

However, the literature on AD enhancement technologies explicitly tailored to DAF sludge from various sources, including DWW, is limited. To identify any potential technological options to enhance AD of dairy DAF sludge, this scarcity caused the requirement of broadening the review to include studies on other organic substrates such as DAF sludge from red meat processing (RMP), slaughterhouse facilities, sewage sludge, and waste-activated sludge (WAS) that undergo similar AD processes. This broader approach is essential to identify transferable technologies and gain valuable insights applicable to dairy DAF sludge, as many AD enhancement strategies can be applied across substrates with similar process-level challenges. To systematically evaluate this transferability, this review employs a weighted scoring analysis (WSA) framework that assesses potential technological solutions across technical, operational, economic, and environmental criteria while incorporating substrate relevance as a main part of the confidence level (CL) calculation to reduce bias for the evaluation.

Furthermore, reviews regarding AD for DAF sludge from DWW are also limited. The literature primarily focuses on general substrates, either dairy, lipid-rich wastewater, or FOG [5, 27]. A few studies reviewed DAF sludge, but their focus was mainly on the DAF process and utilization of DAF sludge, rather than the AD process. For instance, Munoz et al. [20] highlight DAF waste's potential as a biogas feedstock when integrated with biological and physicochemical treatments. Bella et al. [5] emphasize pretreatment technologies and anaerobic co-digestion (ACoD) to optimize AD performance, which is relevant to DAF sludge. Salama et al. [6] noted that FOG is an excellent substrate for enhancing biomethane production through co-digestion, as it yields 250–350 % more methane per g volatile solids (VS) compared to the wastewater sludge typically used in municipal digesters. The increase highlights the higher theoretical yield of lipids (1.014 L CH_4 g VS^{-1} compared to 0.415–0.496 L CH_4 g VS^{-1} for carbohydrates/proteins) [28, 29]. This suggests that since DAF sludge is classified as FOG, ACoD offers promising solutions to address challenges such as long-chain fatty acid (LCFA) inhibition and sludge flotation, which apply to DAF sludge. A review of biotechnological waste management also explored biogas production methods that apply to DAF sludge [30].

A comparison of the existing review studies relevant to the AD of DAF sludge is summarized in Tab. 1. This table highlights the scope, key findings, and relevance of prior reviews, identifying critical gaps in knowledge and reinforcing the need for a focused assessment of enhancement strategies for the AD of DAF sludge.

The increasing production of DAF sludge across the dairy industry and the environmental challenges associated with its disposal underscore the importance of sustainable treatment technologies, including AD. However, the performance of this process in treating DAF sludge remains underexplored and limited, requiring the development of technological enhancements to optimize its conversion into biogas. This study evaluated possible enhancement options to provide more effective strategies for overcoming these treatment challenges.

2 Dissolved Air Flotation Sludge

In dairy processing plants, DAF sludge is a major component of the manufacturing process waste. But, the production of DAF sludge from dairy industries can vary based on the specific operational parameters of the dairy processing plant, including the volume of wastewater treated and the concentration of FOGs. For example, Australian sector data indicate that DAF sludge constitutes 48 % of manufacturing process wastes, totaling 50 000 t per year (wet) across processing sites, with management pathways, including composting and valorization options [32, 26]. Considering the significant quantity of waste generated and the requirement to treat it, it is essential to understand the sludge composition for effective treatment and resource recovery.

A comprehensive overview of the primary characteristics of DAF sludge obtained from DWW is presented in Tab. 2. It presents various water quality parameters from different studies, highlighting both the characteristics of DAF sludge and the variability across different sources or treatments. As can be seen, DAF sludge exhibits high COD values ranging from 88.4 to 203.7 g L^{-1}, which is significantly higher than other typical HSW, such as distillery wastewater (40–140 g L^{-1}) [33] and cream cheese whey (80–90 g L^{-1}) [34]. These high-level COD values reflect significant organic matter content, primarily from lipids, carbohydrates, and proteins, making DAF sludge a promising resource for energy recovery through AD [5, 13]. Girault et al. [35] quantified this sludge's composition as 11.4 % VS protein, 50.1 % VS lipid, and 37.5 % VS carbohydrates.

According to Tab. 2, most samples were slightly acidic, as the pH in the inlet of the DAF system was typically adjusted to approximately 4.2 to facilitate the removal of the protein fraction from the DWW. Ammonium concentrations have been reported to be only 5 [36] and 100 mg N L^{-1} [35]. The sludge's carbon-to-nitrogen (C/N) ratio ranged from 41.6 to 50 [35, 37], which is higher than that suggested (20–30) [35, 39]. This indicates a high

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Table 1. Summary of related review paper for AD process using DAF sludge from dairy wastewater.

Refs.	Focus of review	Review highlights	Relevance to DAF sludge AD
[20]	• DAF process • Utilization of DAF sludge	• Wide applicability of DAF process in agriculture, industry, and municipal • DAF performs better when integrated with biological and physicochemical treatment methods • Potential of DAF sludge resource recovery for biogas or biodiesel production	DAF sludge is mentioned, but only a limited discussion about AD
[5]	• Effects of critical parameters on the successful operation of AD to treat DWW • The need for pretreatment technologies and ACoD	• Optimization DWW in AD by considering operational parameters, managing seasonal variability, and exploring pretreatment and ACoD • Potential for recovering intermediate products • Environmental and energy benefits of DWW conversion using AD	Valuable insights applicable to the AD process of DAF sludge: the need for pretreatment and the potential for ACoD
[6]	• FOG use in AD • AD challenges and solutions	• ACoD of FOG can increase biomethane production by 250–350 % (in mL CH_4 g VS) compared to the wastewater sludge • Challenges include LCFAs inhibition, washout, scum formation, and sludge flotation • Understanding microorganisms, interactions, and degradation kinetics can optimize digestion and prevent operational issues • Future research should emphasize characterization, FOG degradation pathways, operational conditions, economic feasibility, inhibition investigations, and improved pretreatment technologies	The sludge's nature with high content of FOG presents the challenges, solutions, and insights discussed in the review, which are also applicable to the AD of DAF sludge
[30]	• Various DWW management approaches • Highlights growing interest in biotechnology for waste conversion into valuable resources • Examines physicochemical and biotechnological solutions	• Methods to treat DWW: physicochemical, biological, and biotechnological techniques • Biotechnological processes convert waste into biogas, bioethanol, bio-hydrogen, and microbial fuel cells • Sustainable bio-refinery facilities can reduce waste disposal burdens and extract valuable bioactive compounds	The discussed treatment methods and challenges are relevant to the AD of DWW
[31]	• FOG can boost biogas production in wastewater treatment facilities • Economic incentives for excess biogas use • Challenges occurred in AD of FOG	• FOG co-digestion increases biogas production by combining municipal biosolids, offering economic and environmental benefits • Problems with substrate and product transportation, sludge flotation, digester foaming, and gas collection system challenges • Further research is needed to understand these issues and optimize grease abatement device policies	DAF sludge and grease trap waste (GTW) share similarities regarding lipids being the primary substance. The insights and research findings about ACoD of FOG with municipal biosolids are relevant for AD of DAF sludge

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Table 2. Reported characteristics of DAF sludge from DWW.

Parameter	References					
	[16]	[36]	[37]	[12]	[35]	[38]
pH	6.7	7.0 ± 0.1	NR	NR	5.5	5.3
Total COD [g L^{-1}]	88.4 ± 1.1	86.0 ± 0.8	NR	188.3 ± 44.9	195.7	203.7 ± 12
Soluble COD [g L^{-1}]	4.6 ± 0.2	NR	NR	NR	NR	NR
TS [%]	4.48 ± 0.2	3.1 ± 0.05	7.8 ± 0.3	6.8 ± 1.4	9.58	16.7 ± 2.4
VS [%]	3.8	2.4	6.8 ± 0.3	6.5 ± 1.4	8.6	NR
Ammonium [mg L^{-1}]	NR	168.5 ± 1.8	NR	NR	100	NR
Sulfate [mg L^{-1}]	NR	11.5 ± 0.5	NR	NR	NR	NR
C:N ratio	NR	NR	50 ± 1.4	NR	41.6	NR
Nitrogen [g L^{-1}]	NR	NR	NR	~0.1	1.9	6.4 ± 0.5

carbon content relative to nitrogen, potentially leading to nutrient imbalances that could inhibit microbial growth and reduce digestion efficiency.

Therefore, due to strict waste treatment regulations, proper treatment of DAF sludge is essential. However, a comprehensive review of global guidelines for managing DAF sludge is still lacking. In milk-producing countries, regulations typically follow a combination of environmental, industry-specific, and local guidelines, as shown in Tab. 3.

As shown, all countries listed (Tab. 3) enforce strict regulations to ensure compliance with their DWW regulations. In New Zealand, DAF solids are permitted for animal feed under the ACVM regulations [13, 41], whereas the USA regulates DWW through the Clean Water Act.

Dairy industries can leverage DAF sludge for AD to enhance sustainability and compliance by utilizing its rich organic content. This approach boosts the operational efficiency, environmental outcomes, and recovery, turning waste into a valuable resource. Despite the treatment and disposal challenges associated with the sludge produced by the DAF treatment system, DAF technology often serves as a crucial pretreatment step before secondary treatment, such as an activated sludge system, which often includes the AD process [20, 45–47].

Table 3. Summary of guidelines and regulations about DAF-derived waste process in different countries.

Country	Guidelines and regulations about DAF-derived waste	Refs.
China	• No specific regulations for DAF systems in DWW treatment • Environmental Protection Law of China mandates pollution prevention and control • Discharge Standards of Water Pollutants for Dairy Industry (GB 21909-2008) guides treatment processes	[40]
New Zealand	• DAF solids from DWW can be used in animal feed under Agricultural Compounds and Veterinary Medicines (ACVM) Regulations, which govern animal feed manufacture and sale • Strict effluent management regulations categorize contaminant concentrations • Ministry of Primary Industries develops a Code of Practice for dairy farm design and operation	[13, 41]
Australia	• Commit to water conservation and recycling in milk processing • Adopt advanced biological treatment systems • Effluent from DAF processes must adhere to local, state, or national water quality regulations	[1]
USA	• EPA regulates DWW under the Clean Water Act, but regional variations in practices vary by state and county • Large dairy operations, classified as Concentrated Animal Feeding Operations (CAFOs), have specific regulations • AD and composting are used for DAF sludge and soil amendments • Land application is subject to EPA regulations on biosolids	[42]
EU countries	• EU countries have transposed EU directives into national legislation, enforcing compliance through specific regulations • Dairy industry uses DAF systems to treat wastewater to meet regulatory requirements and discharge limits. DAF systems are integral to achieving compliance with EU and national environmental standards • EU Regulations: Urban Wastewater Treatment Directive (91/271/EEC) and Water Framework Directive (2000/60/EC) mandate wastewater collection and treatment in urban areas and industrial sectors	[43]
India	• Central Pollution Control Board (CPCB) provides guidelines for the environmental management of dairy farms and gaushalas • Bureau of Indian Standards (BIS) develops standards like IS: 8682:977 for treating and disposing dairy industry effluents	[44]

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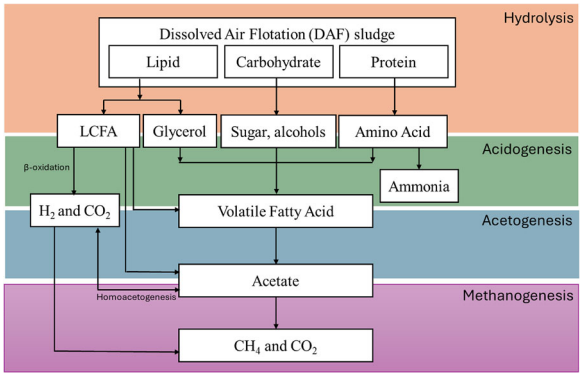


Figure 1. Schematic AD process of DAF sludge.

3 Anaerobic Digestion of DAF Sludge

AD involves various synergistic microorganisms, breaking down complex organic matter without oxygen into biogas, primarily CH_4 and CO_2 [5, 48]. This multistage process consists of hydrolytic, acidogenic, acetogenic, and methanogenic processes (Fig. 1). In hydrolysis, organic substances are broken down into simpler molecules. Subsequently, acidogenesis entails microorganisms decomposing hydrolyzed molecules into organic acids, yielding acetate, CO_2, hydrogen, and inorganic substances such as ammonia and hydrogen sulfide. Methanogenesis subsequently transforms those acids into CH_4 and CO_2 [49, 50].

The AD process of DAF sludge follows all described metabolic pathways, especially for lipids (Fig. 1). Neutral lipids first undergo hydrolysis to form free LCFA and glycerol. Acetogenic bacteria then convert LCFA into acetate and hydrogen via β-oxidation, leading to CH_4 production by methanogens [51, 52].

Additionally, lipid content in DAF sludge has a higher theoretical CH_4 than carbohydrates and proteins, producing approximately 1.3 L of CH_4 per gram of VS [53]. This high CH_4 yield makes it commercially attractive for sustainable energy production. Converting DAF sludge into biogas also provides supplementary environmental and economic benefits, such as reduced sludge volume, lower disposal costs, and nutrient-rich organic fertilizer production for agricultural use [52].

However, because of the characteristics of DAF sludge, the complex microbial processes in AD may encounter challenges (Fig. 2) when it is used as a substrate. The challenges encountered in the AD of this DAF waste can be categorized based on their underlying nature into composition-related, kinetic, and microbial process efficiency, as well as physical, operational, and process-specific challenges. This approach helps to understand the root causes of these problems and how they originate from the inherent properties of the feedstock.

Composition-related issues arise from the substrate's specific characteristics, such as lipid digestibility problems where lipids are hydrolyzed more slowly than carbohydrates, leading to potentially reduced biogas production and yields [6, 31].

Lipids are large hydrophobic molecules poorly soluble in water, making hydrolysis into glycerol and LCFAs a slow and often rate-limiting step in AD [6, 54, 55]. Accumulated LCFAs can inhibit microbial activity by adsorbing onto microbial cells, disrupting mass transfer and cellular functions [56]. This inhibition leads to decreased CH_4 production and process imbalance [51]. Furthermore, overproduction of volatile fatty acids (VFAs) can lower pH and inhibit microbial activity [57, 58].

Nutrient and chemical balance concerns, such as insufficient nutrients, trace metal deficiency, and imbalanced C/N ratios, complicate the process and reduce CH_4 yields and production rates [6, 27]. Insufficient nutrients can hinder microbial activity, slowing digestion. Trace metal deficiencies impair key enzymes in methanogenesis, reducing CH_4 yields [59]. However, high C/N ratios can cause acid accumulation and microbial growth inhibition [39].

The AD efficiency is primarily determined by the rates of key biochemical reactions, especially hydrolysis and methanogenesis. Slow rate in these processes increases system failure risk [56, 60]. Slow hydrolysis of complex organic matter causes delayed and incomplete digestion, leaving increased residual organic matter in the effluent [52]. A balanced microbial population is essential for efficient methanogenesis, as imbalances can significantly affect CH_4 production. Methanogenic archaea, the key microorganisms in CH_4 production, are highly sensitive to environmental changes such as pH fluctuations and inhibitors such as ammonia. Deviations from optimal temperatures can reduce



Figure 2. Challenges of the AD process of DAF sludge according to their underlying nature.

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Table 4. The level of severity and difficulty of challenges for the AD process of DAF sludge.

Challenge category	Challenge	Severity	Difficulty	Refs.
• Lipid digestibility problems and low hydrolysis rate leading to LCFA accumulation	• Lipid digestibility • Low hydrolysis • LCFA inhibition	Very high	High	[53, 55, 57, 58, 63]
• Inhibition leading to low methanogenesis rates (ammonia and VFAs)	• Low methanogenesis rate • Ammonia and VFA accumulation	Very High	High	[67, 68]
• Feed variability	• Hydraulic and nutrient shock	High	Moderate	[5, 27]
• Nutrient and chemical balance causing microbial imbalance	• C/N ratio • Trace metal • Microbial imbalance	High	Moderate	[69–71]
• Operational issues due to process instability	• Flotation • Foaming • Clogging	Moderate	Low to moderate	[72]
• Odor		Moderate	Low to moderate	[73]

CH_4 generation and disrupt the process, potentially leading to system failure [61, 62].

Physical challenges include poor mixing, foaming, clogging, flotation, and variable wastewater composition. Foaming, induced by surfactants and gas production rates, can cause operational issues such as overflow, blockage, and equipment damage [6, 63]. Clogging results from solid accumulation, LCFA precipitation, and biofilm formation. Sludge flotation occurs when biogas bubbles attach to sludge particles, reducing their density and hindering solid–liquid separation. This can result in biomass loss and higher downstream treatment costs [64]. Foaming can cause overflow, blockage, and damage to equipment.

Odors from the degradation of sulfur-containing amino acids, ammonia emissions, and volatile organic compounds (VOCs) require mitigation measures due to their potential for corrosion. Dewaterability is challenging because of the high lipid content that forms fine colloidal particles that are difficult to separate. Microbial production of extracellular polymeric substances (EPS) also increases sludge viscosity, making dewatering less effective [65]. Consequently, disposal costs rise, and mechanical dewatering consumes more energy.

Among the identified problems, some challenges can be addressed with relatively simple solutions, such as adding micronutrients to counteract trace element deficiencies or enhancing mixing techniques to minimize physical issues like floating. However, the more critical issues that significantly compromise the process should take precedence. Therefore, it is crucial to understand the severity and complexity of these challenges (Fig. 2) to effectively prioritize issues and allocate resources [27, 66]. Severity denotes the challenge's impact on the AD process, where higher severity disrupts performance or reduces efficiency more significantly. Difficulty pertains to the complexity of resolving the challenge, with higher difficulty requiring more effort and resources.

Addressing the most critical problems allows dairy industries to maximize the impact of their DAF sludge treatment solutions. In Tab. 4, these challenges are organized based on their interconnections, significance, and intricacy. This table highlights lipid digestibility issues and low hydrolysis rates, leading to LCFA accumulation, which are the most critical challenges, with very high severity and difficulty, as they can most likely inhibit AD performance.

Inhibition of methanogenesis due to ammonia and VFA buildup is also rated as very high in severity and difficulty, threatening CH_4 and biogas yield. Variability in DWW composition has high severity and moderate difficulty that can complicate consistent AD operations. Nutrient imbalances, such as C/N ratio and trace metal deficiencies, also have high severity but moderate difficulty that can affect microbial stability. Operational challenges such as foaming, clogging, and odor issues are rated as moderate in severity and low to moderate in difficulty, posing less risk but still requiring management. These challenges go beyond mere obstacles; they are core benchmarks for determining the relevance and effectiveness of any AD technology for DAF sludge. A viable solution must demonstrably confront and resolve these critical issues, as this capacity often underpins its broader acceptance and long-term success. Consequently, any promising approach to improving AD of DAF sludge should be assessed on its ability to address these challenges effectively.

4 Potential Technological Solutions to Enhance the AD of DAF Sludge

Literature specifically addressing the AD of DAF sludge is limited. To our knowledge, only a limited number of technological solutions have been implemented to convert dairy DAF sludge into biogas. These include pretreatment [16, 74], ACoD [35], and the incorporation of additives [11].

However, studies on other substrates suggest that some technological solutions could potentially address the challenges of the AD process of DAF sludge [31]. For instance, pretreatment technologies such as alkaline pretreatment have shown an increase in the digestibility of organic components in DAF sludge from slaughterhouse wastewater (SHW) [16]. ACoD, which involves mixing this sludge with other substances such as WAS, can help maintain a proper nutrient balance and mitigate inhibition effects. Different technological solutions, such

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Table 5. Summary of existing pretreatment technologies for AD enhancement (expressed by percentage CH_4 increased production) of DAF sludge.

Pretreatment	Source of DAF sludge	Details of the pretreatment	AD setup		AD enhancement in CH_4 yield increase [%]	Refs.
			Mode	Scale		
Acid	SHW	6 M HCl, 2 %–8 % w/v of HCL in the substrate, pH 2–2.5, 4 h	Batch	Lab	2.9	[78]
Alkaline	RMP	NaOH 7 g L^{-1}, 24 h	Batch	Lab	8.5	[79]
Alkaline	SHW	2 M NaOH, 6 %–14 % w/v of NaOH in the substrate, pH 12–12.2, 4 h	Batch	Lab	14.7	[78]
Enzymatic	SHW	Liquid Certizyme 5, 60 mg L^{-1}, 24 h	Batch	Lab	2.9	[78]
Thermal	RMP	Thermobaric autoclaving at 121 °C 30 min, followed by 24 h of cooling	Batch	Lab	3.3	[79]
Thermal	SHW	70 °C, 60 min	Batch	Lab	5.9	[78]
Thermochemical	RMP	NaOH 7 g L^{-1} 24 h, 121 °C 30 min	Batch	Lab	8.3	[79]
Ultrasonic	DWW	20 kHz, 300 W power; pulsed and continuous modes; 30 kJ g^{-1} TS	Batch	Lab	36.0	[16]
Ultrasonic	SHW	24 kHz, 5.6 ± 0.3 kJ g^{-1} TS	Batch	Lab	8.8	[78]
Hydrodynamic cavitation	DWW	Vortex-based HC device, 20 L min^{-1}, 250 kPa pressure drop	Batch	Lab	82	[74]

as additives, including bentonite and activated carbon, have proven helpful in reducing blockages and improving microbial activity [75, 76]. Moreover, modifying operational parameters such as temperature and reactor configuration can also be crucial for stabilizing digestion and maximizing biogas production [77].

4.1 Pretreatment Technologies

Pretreatment processes can enhance the biodegradability of the substrate before the AD process. The improvement occurs by breaking down complex organic matter, increasing nutrient availability for microbes, and aiding biological degradation.

As indicated in Tab. 5, only two pretreatment technologies have been investigated to enhance dairy DAF sludge within the AD process. In contrast, the majority of DAF sludge studies are from RMP and SHW. Furthermore, these investigations are predominantly conducted at a laboratory scale in batch mode, highlighting the early stage of pretreatment research for this type of sludge.

As seen from Tab. 5, pretreatment methods for enhancing AD of DAF sludge include acid, alkaline, enzymatic, thermal, thermochemical, ultrasonic, and hydrothermal. Both ultrasonic and hydrothermal demonstrate the highest potential for boosting CH_4 production, with an increase of 82 % and 36 %, respectively [74]. Alkaline pretreatment has also shown promise, yielding a 14.7 % increase in biogas output. Enzymatic pretreatment, while more specific in its action, provided a relatively lower CH_4 yield enhancement of 2.9 % using Certizyme 5 [78]. Thermal pretreatments, by thermobaric autoclaving and heating to 70 °C, improved CH_4 production by 3.3 % and 5.9 %, respectively. This reflects the potential of thermal processes to moderately increase AD efficiency. Thermochemical processes, by applying NaOH in combination with thermobaric autoclaving, led to an 8.3 % CH_4 yield increase.

The existing literature on the enhancement of AD for dairy DAF sludge is limited, as mentioned. Furthermore, this review seeks to derive insights from other studies concerning technological options for AD enhancement, including pretreatment. Accordingly, this research synthesis compares and evaluates the identified pretreatment technologies to assess their applicability to dairy DAF sludge. Later, the identified pretreatment was assessed based on predefined similarity criteria, which are integrated into the CL calculation within the WSA framework.

4.1.1 Acid Pretreatments

Acid pretreatment hydrolyzes complex molecules by donating protons, breaking ester bonds in lipids, denaturing proteins, and converting polysaccharides into simpler sugars [80]. Its severity varies with acid concentration and temperature, disrupting cell walls and increasing substrate availability for microorganisms. It could help emulsify lipids and break down triglycerides into free fatty acids (FFAs) in DAF sludge, which can be readily utilized in AD [80]. However, excessive FFAs can lead to inhibition of microbial activity, especially methanogens.

Tab. 6 shows the impact of pretreatment for AD enhancement ranges from 14 % to 59 %, depending on the substrate, conditions, and reactor system. This table suggests moderate biogas enhancement with acid pretreatment for other sludge that could be observed for DAF sludge, provided the right conditions are applied.

The mechanism of ester bond hydrolysis and lipid emulsification is particularly relevant for dairy DAF sludge's high fat content, suggesting moderate-to-high transferability potential from WAS and sewage sludge applications.

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Table 6. Performance of acid pretreatment for AD enhancement (expressed as increased biogas or CH_4 yield) of various sludges.

Substrate	Reagent	Biogas/CH_4 enhancement	Refs.
WAS	0.2 mol L^{-1} formic acid	59 % (CH_4)	[81]
Sewage sludge	Peracetic acid (10 % PAA, 14 % H_2O_2, 25 % acetic acid by weight) 2 m mg^{-1} VS PAA	40 % (Biogas)	[82]
WAS	HCl until pH 2	14 % (Biogas)	[83]
WAS	Free nitrous acid (HNO_2) 1.8 mg N L^{-1}	16 % (CH_4)	[84]
WAS	Formic acid 1.1 mol L^{-1}	59 % (CH_4)	[85]

4.1.2 Alkaline Pretreatments

The primary mechanism of alkaline pretreatment for lipid-rich wastewater is saponification, where alkali agents break ester bonds in complex organic molecules, facilitating their breakdown into smaller, more biodegradable components. Alkaline pretreatment typically uses reagents, for example, NaOH, KOH, and lime ($Ca(OH)_2$) at mild temperatures and pH levels above 7. As seen in Tab. 7, alkali pretreatment can improve AD by 13–100 %. Alkaline pretreatment of WAS with NaOH 7 g L^{-1} for 30 min led to a 13 % increase in CH_4 production. Lime can also improve CH_4 production, as seen in the study by Vlyssides and Karlis [86] where CH_4 yield increased by 30 % after pretreatment at temperatures between 50 °C and 90 °C.

The saponification effect can directly apply to lipid-rich dairy DAF sludge, considering DAF sludge has lipid content that can be degraded using saponification mechanisms. The saponification effect of this pretreatment implies that this method can improve the digestibility of the lipid-rich DAF sludge. Alkaline conditions promote fat emulsification, making it more accessible for microbial degradation [62]. However, the drawbacks include forming inhibitors (soap production), scaling and sludge issues, environmental impact, and cost.

Table 7. Performance of alkaline pretreatment for AD enhancement (expressed as increased biogas or CH_4 yield) of WAS and dung.

Substrate	Reagent	Biogas/CH_4 enhancement	Refs.
WAS	NaOH 7 g L^{-1}, 30 min	13 % (CH_4)	[87]
WAS	KOH 65 mEq dm^{-3}, 170 °C, 15 min	54 % (biogas)	[88]
WAS	H_2O_2 150 mmol L^{-1} $FeSO_4$: 5 mmol L^{-1}	16 % (biogas)	[88]
WAS	Lime, adjusted continuously to maintain pH 8–11 at 50–90 °C	30 % (CH_4)	[86]
Cattle manure	NaOH (1 %), 7 days	100 % (biogas)	[89]

Table 8. Performance of ozonation pretreatment for AD enhancement (expressed as increased biogas or CH_4 yield) of various sludges.

Substrate	Conditions	Biogas/CH_4 enhancement	Refs.
Mixed sludge from WWTP	0.5 g O_3 250 g TS^{-1}	36 % (CH_4)	[90]
Mixed sludge from WWTP	0.1 g O_3 g COD^{-1}	80 % (CH_4)	[91]
WAS	0.05 g O_3 g^{-1}	17 % (biogas)	[92]
WAS	Gas flow rate 1 L min^{-1}, O_3 inlet 30 mg L^{-1}. 0.16 g O_3 g TS^{-1}	25 % (biogas)	[93]
WAS	Gas flow rate 1 L min^{-1}, O_3 inlet 30 mg L^{-1}. 0.1 g O_3 g TS^{-1}	8 % (biogas)	[93]

4.1.3 Oxidation Pretreatments

Oxidation pretreatment can utilize ozone as a powerful oxidizing agent that can be applied to various organic wastes to enhance the AD process. This pretreatment involves bubbling ozone gas through the substrate or exposing it to an ozone-rich atmosphere. It can be used in both dry and aqueous systems. The primary mechanism of this pretreatment is the ozone's reaction to breaking down the structure of the organic material, like proteins and lipids, making it more accessible for microbial digestion [90]. Therefore, this ozone pretreatment also possibly increases the digestibility of DAF sludge, converting its large organic molecules, accelerating hydrolysis, and reducing the waste's viscosity, facilitating easier processing in AD systems and improving microbial contact with organic matter.

As seen in Tab. 8, this ozonation significantly improves AD processes ranging from 8 % to 80 % in mixed sludge and WAS, providing a promising approach to optimizing biogas production in DAF sludge.

Tab. 8 presents that ozonation of mixed sludge with O_3 0.5 g 250 g^{-1} total solids (TS) resulted in a 36 % increase in CH_4 production. Increasing the ozone dose to 0.1 g O_3 g COD^{-1} led to an even higher 80 % CH_4 increase, highlighting the dose-dependent effectiveness of ozonation in enhancing AD. Similarly, WAS treated with a gas flow rate of 1 L min^{-1} ozone inlet at 30 mg L^{-1} showed moderate biogas enhancements, with a 25 % increase at 0.16 g O_3 g TS^{-1} and an 8 % increase at 0.1 g O_3 g TS^{-1}. Nonetheless, ozone pretreatment has drawbacks, such as cost, efficiency concerns, formation of inhibitory byproducts, and challenges related to scalability and equipment.

4.1.4 Microaeration Pretreatment

Microaeration pretreatment can enhance biogas production from organic wastes by adding small amounts of oxygen to an anaerobic digester by stimulating oxidative reactions, breaking down organic materials, and increasing substrate availability for fermentation and methanogenesis [94, 95]. The presence of limited oxygen also aids in the partial oxidation of LCFAs,

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Table 9. Performance of microaeration pretreatment technology for AD enhancement (expressed as increased biogas or CH_4 yield) of various organic wastes.

Substrate	Conditions	Biogas/CH_4 enhancement	Refs.
WAS	Microaerobic, 10 days, 60–70 °C, 1 day	50 % (biogas)	[96]
Food waste	Microaeration 4 days with 37.5 mL O_2 $(L\ day)^{-1}$	21 % (CH_4)	[94]
Corn straw	Microaeration 5–10 mg O_2 L^{-1} per cycle	42 % (biogas)	[97]
WAS	Microaeration 55 °C, 12 h	24 % (biogas)	[98]
Sewage sludge	Air flow rate 0.3 vvm, 48 h, 35 °C	211 % (CH_4)	[95]

converting them into short-chain fatty acids [95]. Microaeration improves mass transfer and mixing, stimulates specific microbial populations, and suppresses sulfide production, reducing toxic and corrosive gases produced by sulfate-reducing bacteria.

Tab. 9 illustrates the application of microaeration pretreatment for various substrates, showing biogas volume and CH_4 enhancements ranging from 21 % to 211 %. The microaeration applied can be operated from 12 h to 10 days under various conditions, such as temperature and air flow rate. But the best condition was a study involving sewage sludge with an air flow rate of 0.3 vvm for 48 h at 35 °C, which yielded an impressive 211 % CH_4 increase, indicating the substantial impact of oxygen introduction in enhancing CH_4 production. Microaeration of food waste and corn straw with microbial agents improves the AD process, demonstrating the potential of combining microaeration with microbial augmentation.

Microaeration pretreatment for DAF sludge has potential drawbacks such as over-oxygenation risk, limited scalability, and process complexity. Over-oxygenation may transition the system to aerobic conditions, inhibit methanogenesis, and reduce biogas production. Furthermore, balancing anaerobic and microaerobic conditions complicates the system, requiring advanced monitoring and control.

4.1.5 Enzymatic Pretreatment

Enzymatic pretreatment, involving enzymes like cellulase, lipase, and protease, enhances solubility and accessibility for microorganisms by breaking down ester bonds between glycerol and fatty acids in complex organic molecules into simpler components, i.e., FFAs that can facilitate the AD process [99, 100]. This pretreatment can enhance solubility and accessibility, reduce inhibitory LCFAs, facilitate the uptake of shorter chain fatty acids, and increase lipid particle surface area for microbial enzymes. Furthermore, it can improve mass transfer, stimulate microbial activity, enhance system stability and efficiency, and minimize scum and foaming [100, 101].

The enhancement performance of enzymatic pretreatment is proven by substantial biogas or CH_4 increases observed in WAS, food waste, sewage sludge, and lipid-rich DWW (Tab. 10). The

Table 10. Performance of enzymatic pretreatment for AD enhancement (expressed as increased biogas or CH_4 yield) of various organic wastes.

Substrate	Conditions	Biogas/CH_4 enhancement	Refs.
WAS and food waste	Fungal mash 2 g L^{-1} substrate	46 % (CH_4)	[101]
Sewage sludge	Fungal mash 2 g L^{-1} substrate	53 % (CH_4)	[101]
Food waste	Fungal mash 2 g L^{-1} substrate	34 % (CH_4)	[101]
Lipid-rich DWW	Porcine pancreas lipase 0.5 % w/v and 10 mM Ca^{2+}, pH 8 (1 M NaOH)	58 % (biogas)	[102]
WAS	Protease 1 %/g TSS	26 % (biogas)	[103]

observed increases range from 26 % to 58 %. Yin et al. [101] conducted enzymatic pretreatment on three substrates utilizing an equivalent dose of fungal mash. They observed that this pretreatment resulted in a 46 % increase in CH_4 production for a mixture of food waste and WAS, whereas sewage sludge exhibited a 53 % increase.

The successful application of this pretreatment to various substrates, especially lipid-rich DWW, suggests potential applicability to DAF sludge. However, the limitations of this pretreatment method are high enzyme costs, reduced activity due to changes in pH and temperature, and potential ammonia release from protein degradation, which could inhibit methanogenesis.

4.1.6 Hydrothermal Pretreatment

Thermal pretreatment exposes complex organics to high temperatures, causing hydrolysis and degradation that disrupt cells and solubilize organic matter, enhancing biodegradable component accessibility. Hydrothermal pretreatment (HTP) occurs when water acts as a heat transfer medium and a reactant. Water's thermal conductivity enables rapid heat distribution and efficient conduction, facilitating hydrolysis reactions that break down complex polymers and improve lipid solubility due to its solvent properties [104, 105].

HTP is a well-researched pretreatment technology successfully implemented commercially in a full-scale application with leading providers, including Cambi, HAARSLEV, and Veolia Water Technologies [6]. This pretreatment's remarkable effectiveness at the commercial scale has prompted many studies to explore its potential use in conditioning waste substrates high in FOG, including DAF sludge, in the future [6].

HTP also reduces the viscosity of the sludge, improving its flow characteristics and making it easier to handle during AD [105]. Hydrothermal could help break down fats and other organic substrates, such as proteins and remaining casein in DAF sludge, into simpler compounds, making them more easily digestible by microorganisms [80].

The application of HTP in various substrates, as seen in Tab. 11, ranges in biogas and CH_4 yield increase from 44 % to 260 %. This enhancement can potentially happen for DAF sludge as well.

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Table 11. Performance of HTP for AD enhancement (expressed as increased biogas or CH_4 yield) of various organic wastes.

Substrate	Conditions	Biogas/CH_4 enhancement	Refs.
Sewage sludge	186 °C, 105 min	44 % (CH_4)	[106]
Sewage sludge	180 °C for 76 min	340 % (CH_4)	107]
Pelagic Sargassum	140 °C and 30 min	179 % (biogas)	[108]
Mixture of primary sludge and WAS (55 %:45 %)	145 °C, 30 min	260 % (biogas)	[109]
WAS	175 °C, 30 min	62 % (CH4)	[65]

The fat melting and protein breakdown mechanisms are highly suitable for dairy DAF sludge characteristics, indicating strong transferability potential across all substrate types tested.

The drawbacks of HTP are energy cost due to the high temperatures required, formation of recalcitrant compounds from carbohydrate breakdown, and ammonia release from protein breakdown. These issues must be managed carefully to ensure the successful application of DAF sludge, for example, by optimizing pretreatment conditions, implementing heat recovery systems, and increasing feedstock concentration [104].

4.1.7 Thermochemical Pretreatment

Thermochemical pretreatment is widely recognized as one of the most efficient and extensively studied approaches for FOG [110]. It uses heat and chemical agents like H_2SO_4, NaOH, or H_2O_2 to break down complex organic molecules. Heat disrupts the physical structure, while the chemical agent aids in the solubilization of organic compounds, making them more accessible to microorganisms during AD. This can boost biogas production by 20–83 % (Tab. 12). High-temperature alkali pretreatment led to a 79 % CH4 increase, whereas microwave and H_2O_2 pretreatment also showed effectiveness. Combining these methods can lead to a 20 % CH_4 increase (Tab. 12), highlighting the potential of alkali-driven thermochemical pretreatment for AD performance.

Table 12. Performance of thermochemical pretreatment technology for AD enhancement (expressed as increased biogas or CH_4 yield) of WAS.

Substrate	Conditions	Biogas/CH_4 enhancement	Refs.
WAS	NaOH 130 °C	20 % (biogas)	[112]
WAS	Microwave 600 W, H_2O_2 80 °C, 0.2 g g^{-1} TS, heating rate 20 °C min^{-1}	20 % (CH_4)	[113]
WAS	NaOH 20 mEq L^{-1}, 24 h	83 % (biogas)	[114]
WAS	NaOH 7 g, 121 °C, 30 min	38 % (CH_4)	[87]
WAS	NaOH 7 g L^{-1}, 121 °C, 30 min	79 % (CH_4)	[115]

Table 13. Performance of UP technology for AD enhancement (expressed as increased biogas or CH_4 yield) of various organic wastes.

Substrate	Conditions	Biogas/CH_4 enhancement	Refs.
Thickened mixed sludge	30 500 J g^{-1} TS, 26 kHz, 55 °C at 70 rpm for 13 h	50 % (CH_4)	[121]
Sewage sludge	20 kHz, 200 W, 30 min	138 % (biogas)	[122]
Sewage sludge	353 J g^{-1} TS 15 min in an ice bath, 20 kHz, 50 W	34 % (CH_4)	[123]
WAS	35 000 J g^{-1} TS, 20 kHz, 34 min	31 % (CH_4)	[124]
WAS	108 000 J g^{-1} TS 20 kHz	84 % (biogas)	[125]

Thermochemical pretreatment of lipid-rich substrates like DAF sludge involves thermal melting of fats and chemical hydrolysis of organic complexes. This enhances the hydrolysis phase and lowers sludge viscosity, facilitating easier processing in the digester. However, this pretreatment has drawbacks, such as high energy costs, chemical usage, inhibitory by-product formation, and ammonia release. These energy-intensive processes can elevate operational expenses and complicate chemical disposal or neutralization [88]. The chemical addition can also lead to toxic intermediates that inhibit methanogenesis if not adequately managed.

4.1.8 Ultrasonic Pretreatment

Ultrasonic pretreatment (UP) is a highly effective physical treatment for breaking substrates in the AD reaction by using high-frequency sound waves to disrupt the cell walls of microorganisms and break down complex organic molecules [6, 116, 117]. This wave causes the formation and collapse of microbubbles in the liquid substrate and generates high shear forces that reduce particle size and increase the organic matter solubilization [118–120]. The viability of this pretreatment has been demonstrated at a commercial scale, with successful implementations by companies such as Ultrawaves GmbH and Veolia using their Sonix Technology.

As seen in Tab. 13, the biogas and CH_4 production enhancements from UP can range from 31 % to over 138 %, depending on the substrate, intensity of the treatment, and subsequent AD conditions.

Cell wall disruption and organic matter solubilization mechanisms show universal applicability, suggesting high transferability confidence regardless of substrate origin.

Applying this pretreatment for DAF sludge will have challenges, such as significant capital expenses. Optimizing energy efficiency and reducing installation and maintenance costs are crucial for successfully applying UP.

In addition to UP, hydrodynamic cavitation (HC) operates through a mechanism similar to the degradation of organic substances prior to AD. Both techniques utilize cavitation,

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Table 14. Performance of ACoD (in terms of biogas or CH_4 yield increase) using DAF sludge and other substrates.

Substrates	Ratio	Biogas/CH_4 enhancement	Refs.
DAF sludge from GTW: sewage sludge	77:23 (VS basis)	138 % (CH_4)	[131]
DAF sludge: SHW	1:1 % v/v	18 % (CH_4)	[132]
Pig slurry: sardine oil waste	95:5 % v/v	274 % (CH_4)	[128]
Poultry manure: oil mill wastewater	75:25 % v/v	21 % (CH_4)	[133]
FS DWW: Sewage sludge	COD ratio of FS to SS ranges from 0.4 to 1.9	8 % (CH_4)	[127]
FS pig SHW: municipal sewage sludge	SHW was mixed with sewage sludge in varying ratios, with up to 50 % SHW	45 % (biogas)	[134]
Sewage sludge: GTW	54:46 (VS basis)	66 % (CH_4)	[135]
FS pig SHW (intestines, meat tissue, and bristle): sewage sludge	Ratio 21 %: 50 %: 21 %: 8 %. Ratio of SHW sewage sludge ranges from 30 % to 50 % SHW (% w/w)	140 % (CH_4)	[134]
Sewage sludge: GTW	70:30 (VS basis)	27 % (CH_4)	[136]
WAS: FOG	48:52 (VS basis)	107 % (CH_4)	[35]
Sewage sludge: GTW	77:23 (VS basis)	27 % (CH_4)	[137]
WAS: FOG	64:36 (VS basis)	137 % (CH_4)	[130]
WAS: FOG	34.5:65.5 (VS basis)	318 % (CH_4)	[129]

characterized by the formation and subsequent collapse of microbubbles, to effectuate physical and chemical transformations in substrates. In the case of UP, cavitation bubbles are generated by acoustic waves, whereas HC induces cavitation through pressure fluctuations within liquid flow [126].

Although the above explanation demonstrates promising technologies tested on related substrates, direct studies on dairy processing DAF sludge remain extremely limited (Tab. 5). This scarcity reinforces the value of our technological options transferability assessment approach and highlights priority areas for future experimental validation.

Therefore, pretreatment's potential ability to improve digestibility, especially with further optimization, requires ongoing research. Exploring other sludge and organic waste may reveal new, effective solutions. Although results vary by substrate, improving digestibility justifies exploring novel approaches that could surpass current methods.

4.2 Anaerobic Co-Digestion

ACoD refers to the concurrent digestion of two or more substrates combined in designated ratios to enhance biogas production by facilitating optimal interaction among microorganisms and giving essential nutrients for the digesting process [5, 6]. This technological solution can improve the AD process of DAF sludge due to its ability to optimize the C/N ratio and create optimal conditions for microbial activity [27]. In addition, ACoD can ensure process stability, reduce inhibition from toxic substances through dilution, and maintain balance, ensuring that microbes have sufficient nutrients to grow and metabolize the substrate effectively [54]. This balance is crucial for producing high-quality and -quantity biogas with high CH_4 contents and for consistent and efficient biogas production. Co-digestion can also improve the thermodynamics of the system, potentially enhancing the AD process's overall performance.

The AD enhancement using ACoD demonstrates a positive impact on the performance of DAF sludge. For instance, mono-digestion of DAF sludge in a semicontinuous reactor resulted in an unstable and unsuccessful AD process. However, co-digestion with swine manure improved CH_4 output to 2.19 L day^{-1}, with an increase in CH_4 concentration of 71 % [15]. Co-digestion with sewage sludge can also increase CH_4 content [127].

Further potential co-substrates have been identified to enhance biogas production from DAF sludge and other organic substrates (Tab. 14). These co-substrates include synthetic wastewater, sugarcane juice, cattle slurry, paunch, sewage sludge, intestines, meat tissue, and bristle. In the study by Ferreira [128], pig slurry combined with waste sardine oil at a 95:05 ratio produced a 274 % CH_4 increase, demonstrating the significant potential of co-digesting animal waste products with lipid-rich substrates. Co-digestion of poultry manure and olive oil mill wastewater showed promising results, with a 21 % increase in CH_4 production. Grease trap sludge proved especially effective as a co-substrate with WAS. A 64:36 VS ratio of WAS and FOG enhanced CH_4 production by 137 %, whereas a 34.5:65.5 VS ratio led to an even more significant 318 % increase [129, 130]. This highlights the effectiveness of FOG-rich substrates in boosting AD performance. Additionally, co-digestion of FS from DWW with sewage sludge resulted in an 8 % CH_4 increase. Although this enhancement was modest, it demonstrates the broad applicability of ACoD across various sludge types and organic wastes.

4.3 Additives

Additives are substances or compounds introduced during the AD process to enhance the process's efficiency, stability, and overall performance [39, 75, 76, 138, 139]. Additives support specific biochemical activities, such as syntrophic, metabolic,

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catalytic, and enzymatic activities, which regulate anaerobic efficiency, mitigate inhibitors such as VFAs, ammonia, sulfur, and heavy metals, and enhance process stability by maintaining optimal conditions in the digester and supporting microbial performance through the supply of essential nutrients [139]. Some additives also enhance substrate degradation, making it more accessible to microorganisms, reducing the initial lag phase, and accelerating biogas production. Additionally, certain additives support electron transfer, adsorb inhibitory compounds, and supply essential trace elements, providing crucial micronutrients that may be lacking in the feedstock. They can decrease viscosity and stimulate direct interspecies electron transfer (DIET), where bacteria directly transfer electrons between cells through physical contact, eliminating the need for soluble carriers like hydrogen and allowing bacteria to decompose organic molecules [39, 140].

The collaboration between microbial species enhances electron transfer, enabling them to break down organic molecules they could not decompose independently. With their porous structure, additives improve dewaterability by adsorbing excess water and fine particles, reducing viscosity, and facilitating solids handling. However, the effectiveness of additives depends on their concentration and physical properties, requiring precise optimization. Excessive or inappropriate amounts can lead to toxicity and inhibitory effects [39, 138].

The choice of additive depends on the specific challenges faced in the process and the desired improvements in AD and process stability. Additives can be categorized into several types: conductive materials, trace metals, nanoparticles, biological additives, and minerals. Carbon-based additives like biochar, activated carbon, and graphene promote DIET, facilitating the consumption of VFAs and providing substrates for methanogens in AD [54, 75]. Materials like bentonite and biochar reduce LCFA toxicity, maintain microbial activity, improve mass transfer, and stabilize pH [141]. In contrast, enzymes offer flexibility, as they can be used both as a pretreatment and as an additive added directly in the digester, increasing the efficiency of organic breakdown. Concerning the application of additives for AD using DAF sludge, there is a lack of research examining the usage of a singular additive to increase the AD process of DAF sludge, apart from its application alongside other technological methods [11]. However, some possible additives to enhance the AD process of DAF sludge are summarized in Tab. 15.

As shown in Tab. 15, various additives can significantly enhance the AD process for different substrates. Commonly used additives include activated carbon, bentonite, biochar, carbon cloth, carbon nanotube (CNT), and graphene. Both activated carbon and bentonite can greatly improve biogas and CH_4 production across various substrates. However, the variability in results and dosages indicates that the additive effectiveness highly depends on the specific substrate and conditions. Additionally, supplementation with trace metals like iron, cobalt, and nickel addresses nutrient deficiencies, enriching the system's nutrient profile and enabling the bacterial consortium to efficiently convert organic matter, particularly lipids, into biogas [71].

Tab. 15 presents that biochar significantly boosted CH_4 production, achieving a 45 % increase in ethanol digestion, an 87 % improvement with glucose substrates, and a 579 % rise in CH_4 production with FOG substrates. Carbon cloth enhanced CH_4 production by 140 % with 1-butanol and 59 % with butanol. CNTs displayed variable performance, increasing CH_4 by 16 % with chicken manure and 69 % with an unidentified substrate. Graphene improved CH_4 production by 27 % in coal gasification wastewater and 51 % in sludge. Calcium-based additives have shown great potential, particularly in FOG treatment.

4.4 Process Modification of Digester Operation

Enhancements to the AD process for treating DAF sludge can be achieved through modifications in reactor operation, which are essential for biomass retention and substrate distribution [164]. Process modification refers to altering the existing process without introducing any further processes, chemicals, additives, or substrates. The procedure can be achieved by elevating the temperature (via thermophilic AD [TAD]) and dividing the AD process into several stages (such as a two-stage AD [TSAD]).

4.4.1 Thermophilic Anaerobic Digestion

Temperature plays a vital role in AD process. Generally, a 35–37 °C mesophilic temperature range is appropriate for successful digestive operations of DWW [5]. Although mesophilic AD (MAD) is the most applied configuration, TAD, which operates at higher temperatures between 50 °C and 60 °C, offers several advantages. These include faster hydrolysis reaction rates, increased gas production, higher pathogen removal rates, and a shorter solids retention time than MAD [165]. Despite these advantages, TAD is frequently regarded as a more energy-intensive, less stable process, and higher microbial sensitivity to variations in environmental conditions, including pH and temperature, compared to MAD [165, 166].

Tab. 16 summarizes the studies conducted on process modification involving TAD. The enhancement level of TAD varies significantly (13–45 % for CH_4, 25–33 % for biogas), likely due to differences in substrate composition and OLR. Digestion of dog food at a high OLR (7.4–20.3 g VS (L day)$^{-1}$) resulted in a 45 % increase in CH_4 production, whereas cellulose digestion saw a 37 % CH_4 boost under similar conditions. The OLR varies significantly between substrates, from 1.5 g VS (L day)$^{-1}$ for food waste to 7.4–20.3 g VS (L day)$^{-1}$ for dog food. This variation could impact the comparability of results.

4.4.2 Two-Stage Anaerobic Digestion

Besides TAD, modifying the process by splitting the AD reactor into two stages can also enhance CH_4 generation. Hydrolysis, acidification, acetic acid formation, and methanation (Fig. 1) are AD steps typically conducted concurrently within a single reactor. However, the complex reactions during those stages differ and require distinct microbial strains and operating conditions, so integrating these four stages within a single reactor can reduce the efficiency of microbiological biomass conversion [2, 172, 173]. In addition, single-stage AD (SSAD) encounters challenges, including reduced system stability, accumulation of VFAs, ammonia inhibition, inadequate buffering capacity, generation of harmful intermediates, and management of high

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Table 15. Additives used to enhance the AD process of various organic wastes.

Additives	Substrates	Dosage	AD enhancement	Refs.
Activated carbon	Coal gasification wastewater	10 g L^{-1}	13 % (CH_4)	[142]
Activated carbon	OFMSW	10 g L^{-1}	42 % (biogas)	[39]
Activated carbon	OFMSW	10 g L^{-1}	36 % (biogas)	[39]
Activated carbon	Dairy manure	0.3 wt.% urea, 0.8 wt.% active carbon	48 % (CH_4)	[143]
Activated carbon	Food waste	Tianjin Zhiyuan Reagent Co., Ltd 15 g L^{-1}	41 % (CH_4)	[144]
Bentonite	Synthetic LCFAs	0.5–5.0 $g_{Bentonite}$ L^{-1}	43 % (CH_4)	[145]
Bentonite	Chicken manure	19.91 g of CM (VS) and thermally modified bentonite (300 °C) with a ratio of 0.03, based on TS, were added	41 % (CH_4)	[146]
Bentonite	Dairy manure	0.3 wt.% urea, 0.6 wt.% bentonite	67 % (CH_4)	[143]
Bentonite	Kitchen waste	$Al_2O_3.4(SiO_2)$. H_2O density 2–3 g cm^{-3}	95 % (biogas)	[147]
Bentonite	Solid fatty residues	Bentonite:grease ratio: 1.5 $g_{Bentonite}$ g VS_{grease}^{-1}	2000 % (CH_4)	[141]
Biochar	Citrus peel	35 g L^{-1}	11.51 % (CH_4)	[148]
Biochar	Ethanol	1.25 g L^{-1}	45 % (CH_4)	[149]
Biochar	Glucose	10 g L^{-1}	87 % (CH_4)	[150]
Biochar	Dairy manure	10 g L^{-1}	38 % (CH_4)	[151]
Biochar	FOG	Calcium-modified biochar 2–10 g L^{-1}	579 % (CH_4)	[76]
Carbon cloth	1-Butanol	500 cm^2 L^{-1}	140 % (CH_4)	[152]
Carbon cloth	Leachate	833 cm^2 L^{-1}	30 % (CH_4)	[153]
Carbon cloth	Butanol	500 cm^2 L^{-1}	59 % (CH_4)	[152]
Carbon cloth	Dry OFMSW	1465 cm^2 L^{-1}	10 % (CH_4)	[154]
Carbon cloth	Dog food	1465 cm^2 L^{-1}	13 % (CH_4)	[155]
CNT	Sucrose	1 g L^{-1}	100 % (CH_4)	[156]
CNT	Chicken manure	0.5–6.5 g L^{-1}	16 % (CH_4)	[157]
CNT	Mixture of aloe peel, dairy manure and sewage sludge	0.36 wt.%	69 % (CH_4)	[158]
CNT	Livestock manure	500 mg kg^{-1}	47 % (CH_4)	[159]
CNT	Sludge from a brewery wastewater treatment plant	1400 mg with the OLR 0.5–4 g COD (L day)$^{-1}$	34 % (CH_4)	[160]
Graphene	Ethanol	1 g L^{-1}	25 % (CH_4)	[161]
Graphene	Coal gasification wastewater	0.5 g L^{-1}	27 % (CH_4)	[142]
Graphene	Sewage sludge	30 mg L^{-1} (5 mg g^{-1})	17 % (CH_4)	[162]
Graphene	Sewage sludge	120 mg L^{-1} (20 mg g^{-1})	51 % (CH_4)	[162]
Graphene	Sewage sludge and food waste	50 mg L^{-1} of 25–400 mL substrates	36 % (CH_4)	[163]

Table 16. Improvement of AD of various substrates using the TAD system compared to MAD as the control.

Substrate	AD enhancement	Refs.
Primary raw sludge	33 % (biogas)	[167]
Food waste	13 % (CH_4)	[168]
Dog food	45 % (CH_4)	[169]
Cellulose	37 % (CH_4)	[170]
Wheat straw	25 % (biogas)	[171]

organic loads [174]. To solve these problems, an approach by splitting the stages into two sequential digesters can be relied on, called a TSAD, which aims to minimize inhibition, enhance stability, and ultimately increase the quality and amount of biogas produced. This approach has been used since 1904 [50, 175].

In a TSAD system, the distinct operation of the hydrolytic-acidogenic stage and the methanogenic stage is implemented to address the limitations of an SSAD system. However, one of the main challenges in AD is the hydrolysis step, which acts as the rate-limiting factor for less biodegradable substrates, such as lignocellulosic waste, because this step involves breaking down tough, complex polymers like cellulose, hemicellulose, and lignin, which are not easily degraded by microorganisms [176]. In contrast, methanogenesis becomes the limiting step for

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Table 17. Improvement of AD using TSAD system in various substrates.

Substrate	Condition	AD enhancement	Refs.
Food waste	OLR: 3 g VS (L day)$^{-1}$, in both stages	40 % (CH_4)	[183]
Food waste	OLR: 12 g VS (L day)$^{-1}$ in the first stage, and OLR: 3.1 g VS (L day)$^{-1}$ in the second stage	2 % (CH_4)	[180]
Food waste	OLR: 3.5 kg TVS m^{-3} HRT: 20 days	22 % (CH_4)	[182]
Pretreated poplar wood biomass	pH 7.2 ± 0.2, in the first stage, pH 5.5 ± 0.2 in the second stage	11 % (CH_4)	[176]
Dog food	OLR: 7.4–20.3 g VS (L day)$^{-1}$	7 % (CH_4)	[169]

highly biodegradable substrates like food waste because it will undergo rapid hydrolysis and acidogenesis, leading to the accumulation of VFAs that can cause low pH conditions and inhibit methanogens [172, 177, 178].

When comparing SSAD and TSAD systems, each has distinct advantages and drawbacks. SSAD is simpler to operate, requires lower capital and operational costs, and is suitable for sites with limited space. However, it is less tolerant of high organic loads, has longer hydraulic retention times (HRTs), and is more prone to operational issues like sedimentation and substrate flotation [50, 179]. In contrast, TSAD provides greater process stability, higher organic loading capacities, and more efficient substrate conversion, especially with highly fermentable residues. They can encounter challenges such as acid accumulation during the acidification stage and must carefully synchronize both stages for optimal performance [2, 180]. Despite these benefits, TSAD systems are more expensive to implement [50, 181].

Tab. 17 summarizes the studies conducted on process modification involving TSAD. TSAD can enhance CH_4 production, but the effectiveness varies greatly depending on specific operational conditions and substrates. The wide range of enhancements for food waste (2–40 % CH_4) indicates that optimizing the TSAD is crucial and not straightforward. Food waste shows the highest potential enhancement (40 % CH_4) and the lowest (2 % CH_4). Another study from Micolucci [182] demonstrated a 22 % increase in CH_4 production with food waste at an OLR of 3.5 kg TVS m^{-3} and an HRT of 20 days. Although the performance varied across substrates, such as an 11 % increase for poplar wood biomass and a 7 % increase for dog food, TSAD remains promising for complex waste streams due to its flexibility in handling different substrate characteristics.

5 Comparative Analysis of Technological Solutions

The utilization of DAF sludge as a substrate for AD presents challenges, including compositional concerns related to FOG content and operational difficulties. The issues highlight the need for innovative technological solutions to improve AD operations for this sludge. Technological solutions, including pretreatment, ACoD, additives, and process modification, have demonstrated efficacy for various organic wastes, enhancing DAF sludge digestion, biogas production, and CH_4 content. Therefore, a comparative analysis is crucial to determine the most efficient ways to improve AD processes' performance related to DAF sludge. This study will offer insights into technological solutions' technical, operational, economic, and environmental dimensions.

Table 18. Potential technological solutions to enhance the AD of DAF sludge.

Pretreatment	Co-substrate for ACoD	Additives	Process modification
Acid	Agricultural Biomass	Activated carbon	TAD
Alkaline	Manure	Biochar	TSAD
Microaeration	SHW	Carbon cloth	
Oxidation	Sewage Sludge	Enzymes	
Thermal	WAS	Graphene	
Enzymatic	Food waste	Magnetite	
Thermochemical		Supplementation	
Ultrasonic		BentoniteCNT	

However, given the limited research on AD enhancement technologies specifically for dairy processing DAF sludge, this review uses a broader scope of various organic substrates to employ a systematic technology transferability assessment and comparison to evaluate existing proven technologies in a broader scope of substrates. Therefore, the comparative analysis methodology in this study consists of two key components: (1) identification of AD enhancement technologies proven on various organic substrates, categorizing them by substrate type and similarity to dairy processing DAF sludge; and (2) comparative assessment where each technology is assessed across technical performance, operational feasibility, economic viability, and environmental impact, where the CL is applied, incorporating substrate relevance, and the quality of the sources used in the comparative assessment. The substrate relevance weighting explicitly accounts for the CL in transferring results from the tested substrate to dairy processing DAF sludge applications. This approach enables systematic identification of the most promising technologies for dairy DAF applications while explicitly acknowledging the limitations and uncertainties associated with cross-substrate technology transfer.

As outlined in the previous section, 25 potential technological solutions (Tab. 18) have been identified to enhance the AD process of DAF sludge, derived from the assessment of dairy DAF sludge and other related organic waste. These solutions fall into four categories: pretreatment, ACoD, additives, and process modification. The pretreatment technologies include acid, alkaline, oxidation, thermal, microaeration, enzymatic, thermochemical, and ultrasonic methods. For ACoD, substrates

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such as agricultural or lignocellulosic biomass, manure, SHW, sewage sludge, WAS, and food waste are considered. Additives with significant potential to improve this sludge include activated carbon, biochar, carbon cloth, enzymes, graphene, magnetite, supplementation (Co, Fe, and Ni), bentonite, and CNT. Additionally, TAD and TSAD are options under the process modification category.

Appropriate scientific methods must be applied to assess the most suitable technologies, considering technological, operational, economic, and environmental factors. Several tools or procedures can be used to evaluate the technologies, for example, WSA or weighted sum method (WSM), which is one of the many multi-criteria decision-making methods [185–187]. WSA was selected for this comparative assessment due to its comprehensive nature, which integrates multiple criteria for a thorough evaluation. It assigns relative importance to each factor, ensuring that critical elements significantly influence the final decision. Additionally, its simplicity facilitates ease of implementation and communication, making it an ideal tool for initial screening, especially when numerous options need assessment [186, 187]. Fig. 3 depicts a modified WSA with a key change where a CL is added to enhance the reliability of the primary assessment.

Therefore, this process started by identifying technological options from the literature, selecting applicable criteria and sub-criteria (Supporting Information Tab. S1), assigning and standardizing scores (Supporting Information Tabs. S2–S6), and weighting each criterion to reflect its importance (Supporting Information Tab. S1). Technologies were then ranked based on these results. Scores were standardized using a scoring guide for comparability. Weights highlight the importance of each criterion and sub-criterion, ensuring proper prioritization. The total score for each technology was calculated by multiplying standardized scores by their weights and summing these across all criteria. The technology with the highest overall score was selected. However, in this assessment, there is an additional measure to be performed to add to the level of confidence and reliability of the assessment.

The final score for the overall WSA was calculated using the following equation:

$$\text{Sum of weighted scores} = \sum W_i S_i \quad (1)$$

where W_i is the weight of the ith sub-criterion, and S_i is the score of the ith sub-criterion.

Supporting Information Tab. S7 exemplifies the framework for evaluating the WSA by presenting a matrix assessing various technologies. The matrix is organized by technology (e.g., T1) and evaluated against assigned sub-criteria. Each criterion and sub-criterion has a designated weight indicating its importance and a score reflecting the technology's performance in that area. The values are then multiplied to produce a weighted score for each sub-criterion. The matrix thoroughly analyzes each technology's advantages and drawbacks across multiple dimensions, including AD enhancement. By summing the weighted scores ($W_i S_i$), a comprehensive score is obtained for each technology, facilitating an unbiased assessment of different technical solutions.

5.1 Assessment Criteria and Sub-Criteria

To ensure a comprehensive assessment, the WSA method for evaluating technological solutions involves selecting criteria covering all significant categories, including technical, operational, economic, and environmental considerations. To align with the categories and scoring approach shown in Fig. 4, the technical and operational assessments were divided into two groups based on the characteristics of the technological options (the first group is pretreatment and process modification, whereas the second group is ACoD and additives), whereas for economic and environmental aspects, the criteria were all the same. Pretreatment and process modification technologies were evaluated for AD enhancement, ability to address related challenges, and technological maturity (technology readiness level or TRL). Meanwhile, ACoD and additives were assessed for AD enhancement and the ability to address associated challenges (Fig. 4b). Operational aspects included usability, maintenance, system compatibility, health and safety, and handling and storage for ACoD and additives. Economic assessment considered capital and operational costs, whereas environmental evaluation focused on wastewater production and gas emissions.

Assigning weights to sub-criteria is essential for comprehensively evaluating technological solutions (Supporting Information Tab. S1). A well-defined scoring guide reduces subjectivity and bias, ensuring consistency and reproducibility across all assessments. The guide provides a structured framework for evaluating each criterion and sub-criterion, aiding decision-making by enhancing transparency and clarity. Supporting Information Tabs. S2–S6 detail the scoring criteria for the primary categories, facilitating a thorough evaluation.

5.2 Confidence Level

Although user-friendly and straightforward, the WSA approach often encounters issues validating scoring subjectivity. Ensuring validated data can enhance the reliability of technology selection. Although the WSA process is typically straightforward, integrating variables like CLs can improve decision-making. More importantly, it is crucial to note that the technological options identified for comparative assessment are derived from studies that do not primarily utilize dairy DAF sludge as their primary substrates. Consequently, a correction factor is necessary to accurately reflect the relevance of these studies as the baseline for the comparative analysis framework.

The CL was used to assess the trustworthiness of data, drawing on existing literature for scoring. In the scoring process, CL indicated data reliability, strengthening the dependability of the WSA. This correction factor can help temper subjectivity in the main WSA framework by reducing the influence of less reliable sources or less relevant to dairy DAF sludge. Assessment criteria included quality, determined by factors like a journal's quartile ranking, the authors' h-index, and citation count, as well as relevance, which measured data alignment with ideal conditions such as substrate similarity, experiment scale, and reactor configuration. Regarding substrate similarity, this sub-criterion for the CL emphasizes evidence derived from matrices

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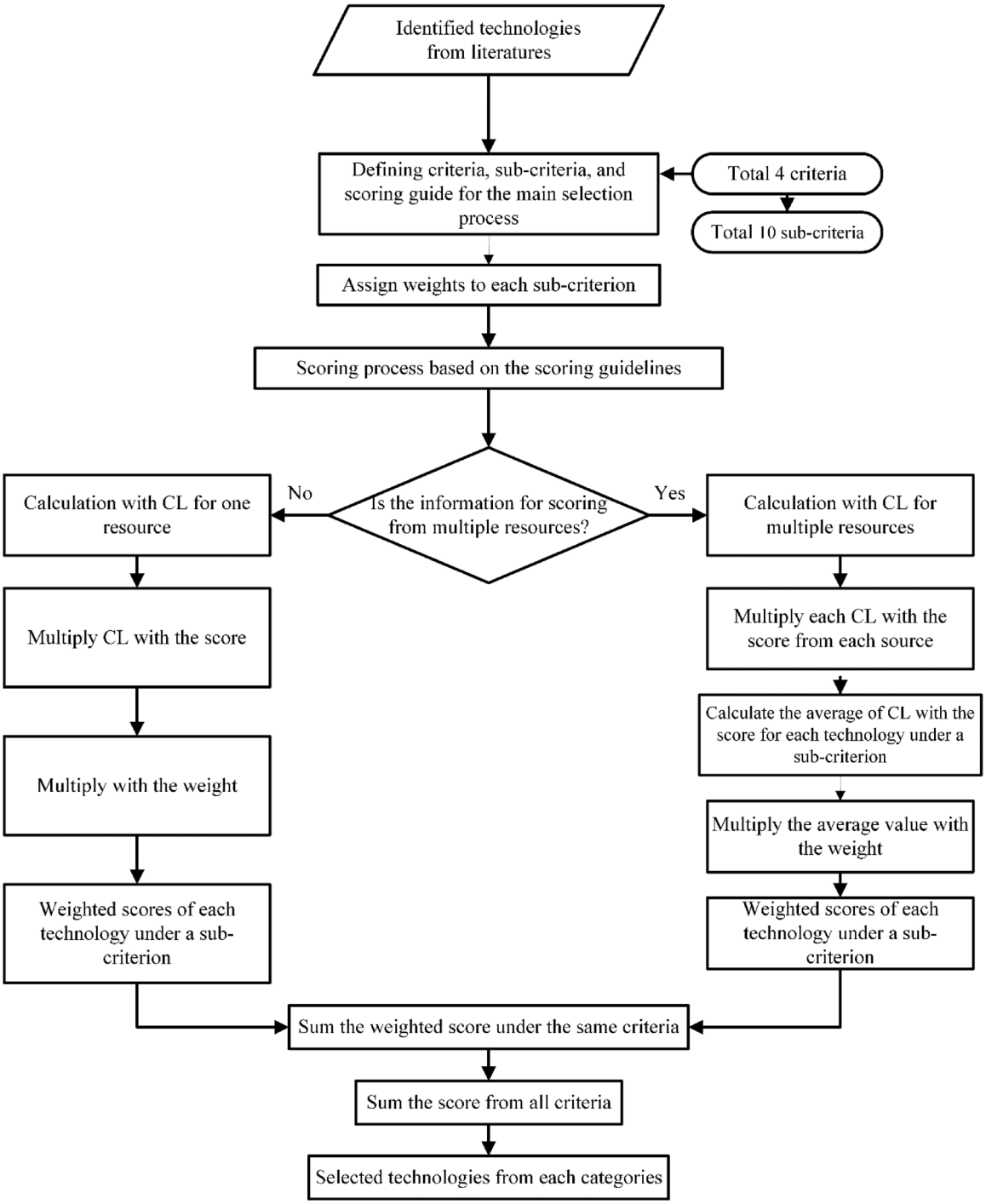


Figure 3. The flow diagram of the modified WSA methodology.

most comparable to dairy DAF sludge. Higher scores indicate a closer resemblance to the target system. The weighting prioritizes directly applicable evidence while also considering insights from related substrates, acknowledging the limited availability of dairy-DAF data. Supporting Information Tab. S8 outlines the criteria, sub-criteria, weights, and scoring guides, ensuring the review's reliability and transparency.

Identifying suitable technological solutions to enhance the AD of DAF sludge and other organic waste highlights the need to incorporate CLs to ensure data comparison accuracy and alignment with desired conditions. For continuous industrial-scale processing, selecting a substrate closely aligned with DAF sludge from DWW is vital, as technology efficacy may not be transferable between substrates. Supporting Information Fig. S1 outlines

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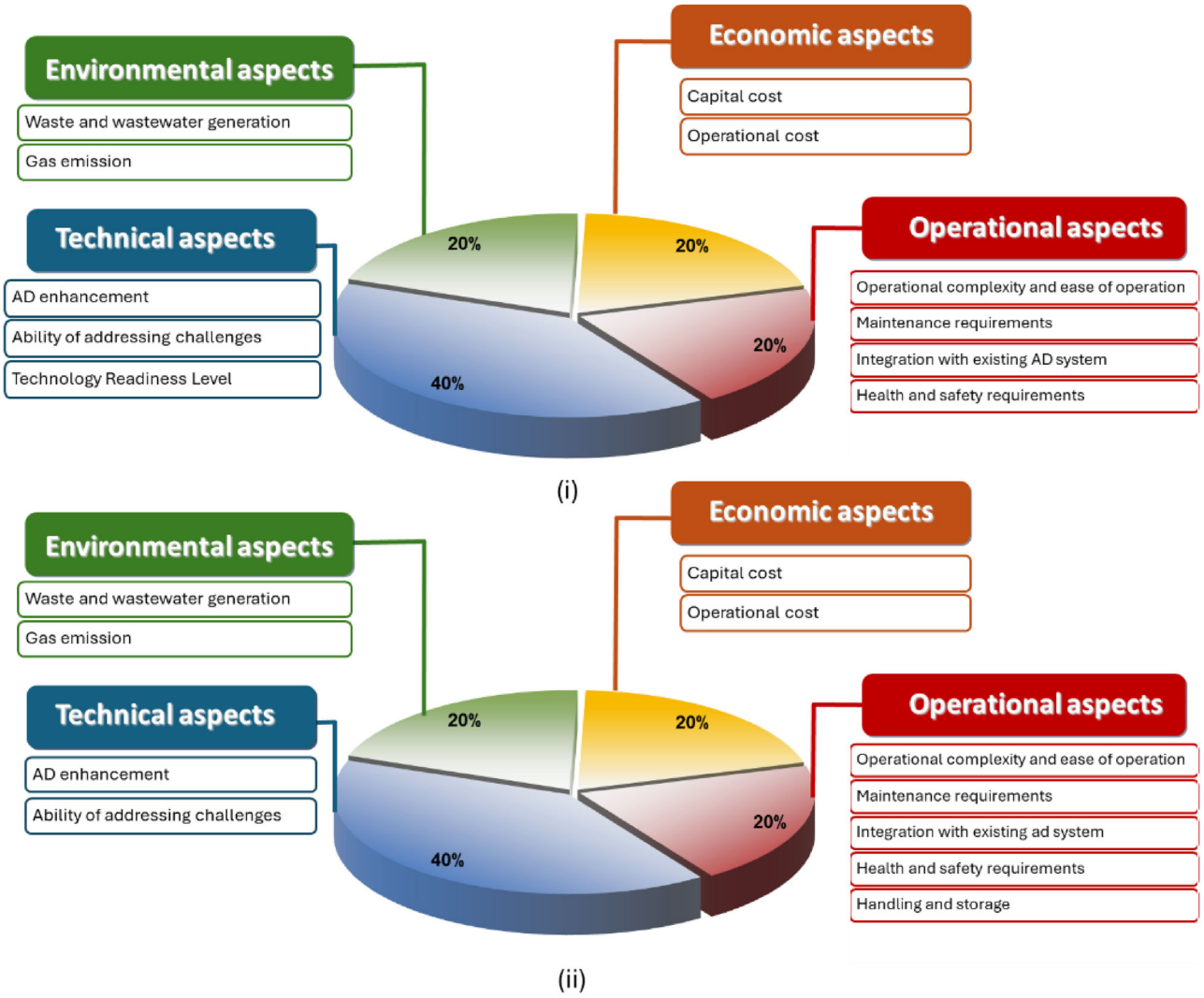


Figure 4. Criteria and sub-criteria used to assess technological solutions: (a) pretreatment and process modification categories, and (b) ACoD and additives categories.

the sequential procedures for calculating the CL to enhance understanding, whereas Supporting Information Tab. S8 provides a matrix illustration. In scoring processes involving a single resource, the score was multiplied by the CL and then by the relative weights assigned to each technology within a sub-criterion, as illustrated by Eq. (2). Supporting Information Tab. S8 shows the updated WSA approach with the corresponding CL:

$$\text{Sum of weighted scores} = \sum W_i s_i \text{CL}_i \tag{2}$$

where W_i is the weight of the ith sub-criterion, s_i is the score of the ith sub-criterion, and CL_i is the confidence level of each resource used to determine the score.

The CL is calculated using the following equation:

$$\text{CL}_i = \frac{\sum \omega_i s_i}{\sum \omega_i s_t} \tag{3}$$

where ω_i is the weight of the ith CL sub-criterion, s_i is the score of the ith CL sub-criterion, and s_i is the ideal (highest) score of the ith CL sub-criterion.

The CL in research evaluates information reliability, with 0 being untrustworthy and 1 being highly reliable. A higher CL signifies better reliability, quality, and relevance, offering a more precise evaluation of future technological applications.

Following the evaluation of the technological solutions in the main assessment using CL, it is critical to identify which technology may efficiently solve the problems faced in the AD process of DAF sludge. This will involve an evaluation of the ability of each technological solution to address the most critical and complex challenges (Tab. 4).

Given the assessed technological solutions for AD of lipid-rich substrates, including DAF sludge, possess unique strengths and limitations, their effectiveness in addressing these challenges can vary significantly. Each technological solution must

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Table 19. Final weighted scoring of technological options with confidence level applied.

		Technical aspects			Operational aspects A: Operational complexity and ease of operation; B: Maintenance requirement; C: Integration with existing AD system; D: Acceptable health and safety requirements; E: Handling and storage					Economical aspects		Environmental aspects	
Type	Technology	Biogas production	Challenges	TRL	A	B	C	D	E	Capex	Opex	Waste and wastewater	Emission
Pretreatment	Acid	0.09	0.07	0.36	0.10	0.08	0.10	0.05		0.26	0.21	0.10	0.21
	Alkaline	0.12	0.38	0.41	0.12	0.12	0.12	0.09		0.35	0.23	0.12	0.23
	Microaeration	0.14	0.49	0.34	0.05	0.11	0.14	0.16		0.33	0.33	0.27	0.27
	Oxidation	0.11	0.29	0.30	0.05	0.05	0.05	0.05		0.05	0.10	0.15	0.15
	Hydrothermal	0.17	0.21	0.46	0.13	0.15	0.15	0.10		0.15	0.25	0.30	0.10
	Enzymatic	0.11	0.18	0.34	0.08	0.08	0.13	0.13		0.30	0.15	0.20	0.30
	Thermochemical	0.13	0.34	0.42	0.02	0.02	0.02	0.02		0.05	0.05	0.05	0.05
	Ultrasonic	0.14	0.25	0.46	0.07	0.05	0.10	0.12		0.20	0.29	0.29	0.29
Co-substrate for ACoD	Agricultural biomass	0.48	0.36		0.05	0.07	0.05	0.12	0.10	0.12	0.18	0.18	0.30
	Manure	0.63	0.61		0.13	0.13	0.10	0.10	0.08	0.32	0.39	0.39	0.19
	SHW	0.97	0.14		0.07	0.04	0.07	0.04	0.04	0.16	0.11	0.16	0.11
	Sewage sludge	0.61	0.36		0.12	0.07	0.12	0.07	0.10	0.30	0.24	0.24	0.18
	WAS	0.80	0.65		0.13	0.13	0.16	0.13	0.10	0.33	0.26	0.26	0.26
	Food waste	0.34	0.34		0.09	0.05	0.09	0.05	0.07	0.23	0.28	0.23	0.17
Additives	Activated carbon	0.53	0.39		0.11	0.09	0.11	0.11	0.11	0.22	0.22	0.22	0.33
	Biochar	0.38	0.44		0.09	0.09	0.09	0.11	0.09	0.22	0.22	0.22	0.27
	Carbon cloth	0.30	0.08		0.07	0.07	0.07	0.12	0.09	0.18	0.18	0.18	0.30
	Enzymes	0.19	0.22		0.10	0.10	0.13	0.08	0.08	0.13	0.13	0.19	0.19
	Graphene	0.23	0.07		0.04	0.04	0.04	0.04	0.04	0.05	0.05	0.11	0.11
	Magnetite	0.24	0.31		0.07	0.07	0.07	0.10	0.07	0.18	0.30	0.30	0.24
	Supplementation	0.31	0.28		0.09	0.14	0.14	0.09	0.11	0.29	0.29	0.29	0.17
	Bentonite	0.78	0.37		0.10	0.08	0.08	0.11	0.11	0.29	0.29	0.29	0.19
	CNT	0.31	0.06		0.02	0.02	0.02	0.02	0.02	0.05	0.05	0.05	0.10
Process modification	TSAD	0.07	0.62	0.44	0.10	0.08	0.13	0.13		0.21	0.26	0.31	0.31
	TAD	0.11	0.36	0.44	0.11	0.08	0.11	0.11		0.21	0.21	0.27	0.21

be carefully assessed based on how well it overcomes specific hurdles in the AD process.

5.3 Evaluation Results

The scoring results for each technological option assessed by using modified WSA are presented in Tab. 19 and Fig. 5. This evaluation emphasizes the importance, applicability, and quality of data, considering the CL for a comprehensive assessment. Tab. 19 displays the final weighted score distribution across the main and sub-criteria for each technological solution, illustrating how individual criterion scores contribute to the overall weighted score. Detailed explanations of each sub-criterion score are presented in Supporting Information Tabs. S12–S17 in the support information offering a thorough justification for the qualitative sub-criteria of pretreatment, ACoD, additives, and process modification.

Overall, according to Fig. 5, the modified WSA identified ACoD as the most promising technological option among the 25 evaluated options, excelling across technical, operational, economic, and environmental criteria. This conclusion is supported by a detailed technical assessment that examined how well each solution addresses the critical challenges in the AD process of HSW, as outlined in Fig. 2 and Tab. 4.

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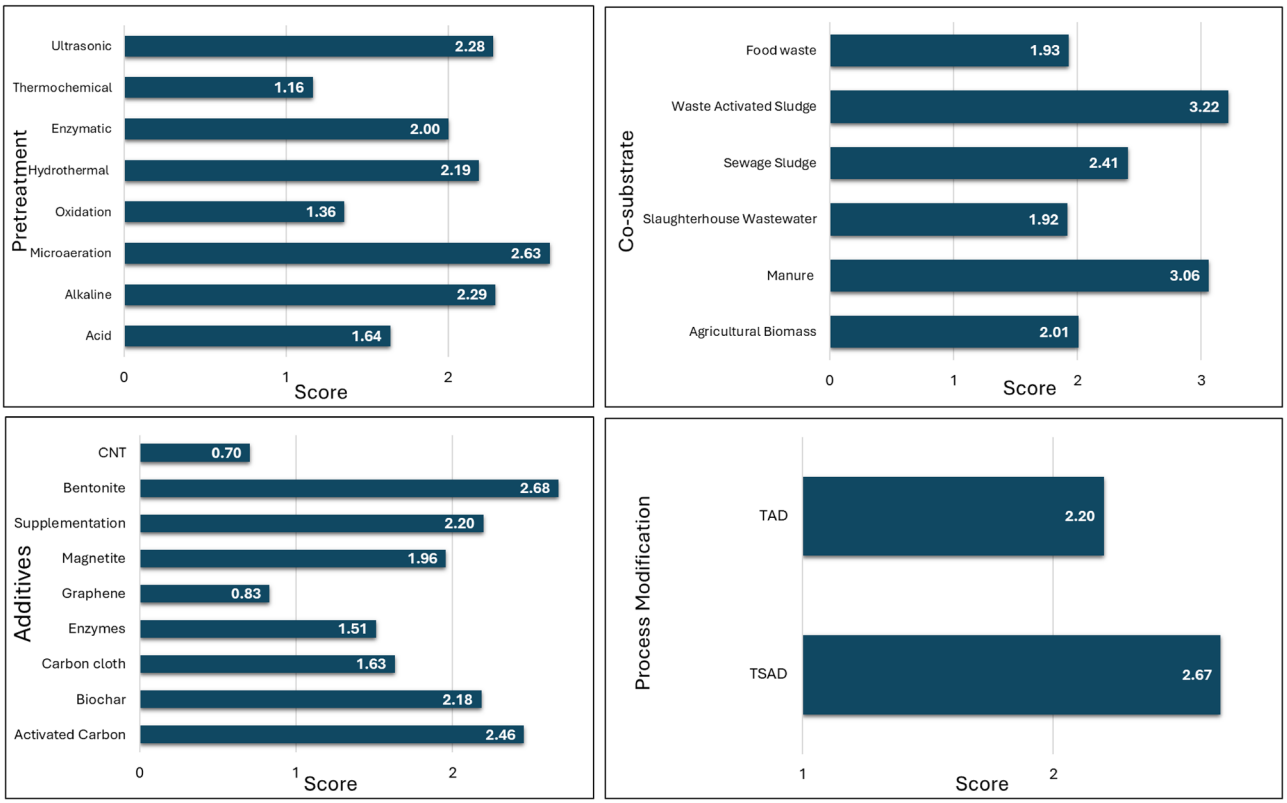


Figure 5. Total weighted score of technological solutions to enhance AD process of DAF sludge.

In the first category, i.e., pretreatment category, Fig. 5 illustrates that microaeration has the highest total weighted score of 2.63, outperforming other evaluated pretreatment options. Microaeration outperformed other pretreatments by balancing all criteria, particularly its ability to enhance biogas production [188, 189]. Moreover, microaeration is viewed as a versatile and relatively low-cost strategy to improve the performance and stability of anaerobic digesters [94, 95]. Meanwhile, enzymatic pretreatments showed moderate effectiveness and practicality. Oxidation and thermochemical pretreatments ranked lowest due to higher costs and technical issues.

The evaluation of various ACoD substrates revealed considerable differences. Dairy WAS scored 3.22, closely followed by manure at 3.06, demonstrating high AD enhancement performance and potential compatibility with DAF sludge. Food waste, however, scored lowest at 1.93 due to issues such as rapid acidification, foaming, and seasonal nutrient fluctuations. Less processed substrates like agricultural biomass, scoring 2.01, face challenges with homogeneity and pretreatment requirements, emphasizing the importance of substrate selection in optimizing ACoD processes.

Among additives, bentonite achieved the highest score of 2.68 due to its cost-effectiveness, high cation exchange capacity, and ability to enhance digestate quality. This naturally occurring clay mineral is considered safe for agricultural use and capable of retaining and releasing nutrients. Bentonite in digestate could improve soil structure, water infiltration, and microbial activity while enhancing soil physical properties like water-holding capacity, thereby increasing biofertilizer value [191, 192]. Activated carbon scored 2.46, offering strong environmental performance but at a higher cost. Biochar excels environmentally and performs well operationally, scoring 2.18. Advanced materials such as graphene and CNT scored the lowest, indicating limited feasibility despite their technical potential.

Process modifications like TSAD tackle systemic challenges within the final category of technological solutions. TSAD surpassed TAD, achieving scores of 2.67 compared to 2.20. Although TAD scored higher in biogas enhancement, owing to its longer research history, TSAD excelled in addressing critical issues such as LCFA inhibition and process instability. By separating hydrolysis/acidogenesis from methanogenesis, TSAD prevents microbial cross-inhibition, facilitating the efficient conversion of lipid-rich substrates (e.g., DAF sludge) into biogas.

Compared to the other technological options categories, especially ACoD and additives, the intricate nature of TSAD's multistage operation may impact aspects like maintenance, monitoring, and control due to the complexities of managing more than one reactor. However, during the assessment, particularly in evaluating the ability of technological options to address key challenges, it became evident that TSAD demonstrated a strong capacity to tackle many of these challenges, particularly in converting substrates such as DAF sludge into biogas.

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Figure 6. Technological solutions and their ability to address the key challenges in AD process of DAF sludge.

This strength is reflected in the WSA-derived scores, though its potential needs further attention, considering its ability to address specific challenges effectively.

Additionally, TSAD has been less explored for four primary reasons: (1) much of the existing research on multistage systems concentrates on bio-hydrogen rather than CH_4 production; (2) TSAD is relatively new, having been developed only in the last 18 years, resulting in a less exploration and literature compared to other technological options; (3) reported improvements in AD performance often indicate a maximum 40 % increase over single-stage systems, leading to fewer direct comparisons with the conventional common system; and (4) AD enhancement encompasses more than just CH_4 yield, including factors such as a greater capacity to handle higher organic loading rates, shorter retention times, and improved stability, which the WSA does not always capture comprehensively due to data limitations for a thorough comparative review of all technological options [2, 173, 175].

As seen in Tab. 19, the primary WSA assessment also included the evaluation of the technological solutions for their ability to address the key problems. At the same time, the scoring of this specific sub-criterion is also depicted in Supporting Information Tab. S18. Fig. 6 is extracted from that part to exhibit that each technological solution may address several difficulties; however, most have limitations in specific areas [2, 50].

As shown in Fig. 6, TSAD demonstrates strengths in most identified challenges, making it a highly comprehensive and versatile technology for AD processes. Similarly, ACoD with WAS successfully tackles methanogenesis inhibition, microbial balance, and operational stability, positioning it as another robust option for enhancing AD efficiency. Other options, such as ACoD with manure and microaeration pretreatment, also display strong potential. Manure's strengths lie in solving lipid digestibility and microbial balance, although it lacks effectiveness in addressing odor issues. Microaeration offers balanced performance across methanogenesis inhibition, microbial balance, and operational-related problems. Thermochemical and bentonite also perform well, making them effective but slightly less comprehensive alternatives.

Integrating challenge-specific analysis into the WSA framework adds critical depth. By pairing WSA's multi-criteria scores incorporating an assessment of how each technology resolves key problems (e.g., LCFA inhibition), stakeholders gain actionable insights into trade-offs between holistic performance and niche capabilities. The WSA serves as a robust foundation for evaluating technical, operational, economic, and environmental factors, whereas the challenge-specific analysis shows strengths and weaknesses that may not be apparent in the overall scores. This combined approach provides a more nuanced view of the capabilities of each technology, resulting in an evaluation

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that is both comprehensive and applicable to real-world scenarios.

6 Conclusion

Treatment of DAF sludge derived from the dairy industry and transferring it into a valuable resource for sustainable renewable energy by applying AD treatment process is facing challenges such as lipid digestibility, low hydrolysis and methanogenesis rate, inhibition (LCFA, VFA, ammonia), imbalance (nutrient, trace element, and microbial), operational, and odor issues. Some technological solutions, including pretreatment, ACoD, additives, and process modifications, were identified to promote the breakdown of difficult organic compounds such as rich lipid content material, balance nutrient profiles, and optimize the AD process.

To evaluate the identified technological solutions, the modified WSA method is used as a tool for comparative analysis by integrating CLs to improve its robust selection process. The technological solutions were scored using technical, operational, economic, and environmental criteria. The modified WSA shows that microaeration pretreatment technology has the most potential in the pretreatment category, WAS has the highest potential co-substrate in the ACoD category, bentonite is rated the best in the additives category, and TSAD is in process modification.

An additional aspect of the primary assessment focused on how different technologies address key challenges in AD enhancement using DAF sludge as the substrate. In this context, TSAD stands out for effectively addressing multiple challenges. It contributes to enhanced lipid digestibility, improved hydrolysis rates, ammonia and VFA inhibition mitigation, better nutrient and microbial balance, and greater operational stability and odor control. While aligned with the WSA evaluation, these strengths offer further insights into TSAD's potential as a versatile solution.

Supporting Information

Supporting information for this article can be found under DOI: https://doi.org/10.1002/cben.70031.



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Abbreviations

AD	Anaerobic digestion
ACoD	Anaerobic co-digestion
BHW	Biomass hydrolysis waste
BOD	Biochemical oxygen demand
CSTR	Continuous stirred tank reactor
CL	Confidence level
DAF	Dissolved air flotation
DIET	Direct interspecies electron transfer
DWW	Dairy wastewater
EPA	Environmental Protection Agency
FS	Flotation sludge
FOG	Fat, oil, and grease
GTW	Grease trap waste
HC	Hydrodynamic cavitation
HRT	Hydraulic retention time
HSW	High-strength waste
HTP	Hydrothermal pretreatment
LCFAs	Long-chain fatty acids

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OLR	Organic loading rate
OFMSW	Organic fraction of municipal solid waste
PS	Primary sludge
RMP	Red meat processing plant
SCOD	Soluble chemical oxygen demand
SS	Sewage sludge
SHW	Slaughterhouse wastewater
SMP	Specific methane production
TAD	Thermophilic anaerobic digestion
tCOD	Total chemical oxygen demand
TOC	Total organic carbon
TN	Total nitrogen
TRL	Technology readiness level
TS	Total solids
TSAD	Two-stage anaerobic digestion
UP	Ultrasonic pretreatment
VFAs	Volatile fatty acids
VS	Volatile solids
WSA	Weighted scoring analysis
WWTPs	Wastewater treatment plants

References

[1] FAO, *Dairy Market Review: Overview of Global Market Developments in 2023*, FAO, Rome **2024**.

[2] P. A. Cremonez, J. G. Teleken, T. R. Weiser Meier, H. J. Alves, *J. Environ. Manage.* **2021**, *281*, 111854. DOI: https://doi.org/10.1016/j.jenvman.2020.111854

[3] A. Kolev Slavov, *Food Technol. Biotechnol.* **2017**, *55* (*1*), 14–28. DOI: https://doi.org/10.17113/ftb.55.01.17.4520

[4] T. J. Britz, C. Van Schalkwyk, Y.-T. Hung, in *Handbook of Industrial and Hazardous Wastes Treatment*, CRC Press, Boca Raton **2004**.

[5] K. Bella, P. V. Rao, *Biomass Convers. Biorefin.* **2023**, *13* (*4*), 2527–2552. DOI: https://doi.org/10.1007/s13399-020-01247-2

[6] E. S. Salama, S. Saha, M. B. Kurade, S. Dev, S. W. Chang, B. H. Jeon, *Prog. Energy Combust. Sci.* **2019**, *70*, 22–42. DOI: https://doi.org/10.1016/J.PECS.2018.08.002

[7] M. Logan, L. C. Tan, P. N. L. Lens, *Int. Biodeterior. Biodegrad.* **2022**, *172*, 105425. DOI: https://doi.org/10.1016/j.ibiod.2022.105425

[8] A. N. Hassan, B. K. Nelson, *J. Dairy Sci.* **2012**, *95* (*11*), 6188–6203. DOI: https://doi.org/10.3168/jds.2012-5732

[9] J. P. Kushwaha, V. C. Srivastava, I. D. Mall, *Crit. Rev. Food Sci. Nutr.* **2011**, *51* (*5*), 442–452. DOI: https://doi.org/10.1080/10408391003663879

[10] M. D. S. Pereira, A. C. Borges, F. F. Heleno, L. F. A. Squillace, L. R. D. Faroni, *J. Clean. Prod.* **2018**, *189*, 729–737. DOI: https://doi.org/10.1016/j.jclepro.2018.04.065

[11] E. Skripsts, L. Mezule, E. Klaucans, *Energies (Basel)* **2022**, *15* (*12*), 4333. DOI: https://doi.org/10.3390/en15124333

[12] R. Lutze, M. Engelhart, *Water Resour. Ind.* **2020**, *23*, 100122. DOI: https://doi.org/10.1016/j.wri.2019.100122

[13] Ministry for Primary Industries, *Recovery of Dissolved Air Flotation Solids at Dairy Premises*, Ministry for Primary Industries, Wellington **2021**.

[14] F. M. Damaceno, E. L. Buligon, J. C. Pires Salcedo Restrepo, M. Chiarelotto, R. K. Niedzialkoski, L. A. de Mendonça Costa, J. de Lucas Junior, M. S. S. de Mendonça Costa, *Sci. Total Environ.* **2019**, *683*, 773–781. DOI: https://doi.org/10.1016/j.scitotenv.2019.05.314

[15] K. S. Creamer, Y. Chen, C. M. Williams, J. J. Cheng, *Bioresour. Technol.* **2010**, *101* (*9*), 3020–3024. DOI: https://doi.org/10.1016/J.BIORTECH.2009.12.029

[16] Y.-C. Liu, S. O'Connor, L. M. Paulo, C. M. Braguglia, M. C. Gagliano, V. O'Flaherty, *Process Saf. Environ. Prot.* **2024**, *183*, 687–697. DOI: https://doi.org/10.1016/j.psep.2024.01.058

[17] C. P. Steele, J. D. West, in *Proc. 1989 Food Process. Waste Conf., Georgia Tech Res. Inst.*, Atlanta, Ga, Vol. 297, Georgia Tech Research Institute, Atlanta **1989**.

[18] E. T. Dennis, M. S. Totzke, in *Food Industry Environmental Conference Proceedings*. Vol. 333, Georgia Tech Research Institute, Atlanta **1991**.

[19] H. W. H. Menkveld, N. C. Boelee, G. O. J. Smith, S. Christian, in *Proceedings of the Water Environment Federation* **2016**, *5* (*9*), 3543–3551.

[20] J. Muñoz-Alegría, E. Muñoz, J. Florez Marulanda, *TecnoLógicas.* **2021**, *24*, e2111. DOI: https://doi.org/10.22430/22565337.2111

[21] M. Quintern, B. Seaton, E. D. Mercer, P. Millichamp, in *Proc. Water. Environ. Fed.*, Water Environment Federation, Virginia **2017**.

[22] Y. Hu, O. Khomenko, W. Shi, Á. Velasco-Sánchez, S. M. Ashekuzzaman, N. Bennegadi-Laurent, K. Daly, O. Fenton, M. G. Healy, J. J. Leahy, P. Sørensen, S. G. Sommer, A. Taghizadeh-Toosi, I. Trinsoutrot-Gattin, *Front. Sustain. Food Syst.* **2021**, *5*, 1–24.

[23] O. V. Okoro, Z. Sun, J. Birch, *J. Clean. Prod.* **2017**, *168*, 1436–1447. DOI: https://doi.org/10.1016/J.JCLEPRO.2017.09.128

[24] E. Atallah, J. Zeaiter, M. N. Ahmad, M. Kwapinska, J. J. Leahy, W. Kwapinski, *J. Environ. Chem. Eng.* **2020**, *8* (*1*), 103599. DOI: https://doi.org/10.1016/J.JECE.2019.103599

[25] K. K. Porterfield, J. Faulkner, E. D. Roy, *ACS Sustain. Chem. Eng.* **2020**, *8* (*4*), 1964–1970. DOI: https://doi.org/10.1021/acssuschemeng.9b06419

[26] T. Canter, T. T. Lim, J. Zulovich, *Nutrient Recovery System for Dairy Farms*, Dissolved Air Flotation and Multi-Disk Press, Missouri **2021**.

[27] A. Abomohra, S. Faisal, R. Ebaid, J. Huang, Q. Wang, M. Elsayed, *Chem. Eng. J.* **2022**, *449*, 137829. DOI: https://doi.org/10.1016/j.cej.2022.137829

[28] Y. Ren, C. Wang, Z. He, Y. Li, Y. Qin, Y.-Y. Li, *Chem. Eng. J.* **2023**, *459*, 141551. DOI: https://doi.org/10.1016/j.cej.2023.141551

[29] R. M. Ziels, A. Karlsson, D. A. C. Beck, J. Ejlertsson, S. S. Yekta, A. Bjorn, H. D. Stensel, B. H. Svensson, *Water Res.* **2016**, *103*, 372–382. DOI: https://doi.org/10.1016/J.WATRES.2016.07.043

[30] M. Kumar Awasthi, A. Paul, V. Kumar, T. Sar, D. Kumar, S. Sarsaiya, H. Liu, Z. Zhang, P. Binod, R. Sindhu, V. Kumar, M. J. Taherzadeh, *Bioresour. Technol.* **2022**, *344*, 126193. DOI: https://doi.org/10.1016/j.biortech.2021.126193

[31] J. H. Long, T. N. Aziz, F. L. D. L. Reyes, J. J. Ducoste, *Process Saf. Environ. Prot.* **2012**, *90* (*3*), 231–245. DOI: https://doi.org/10.1016/J.PSEP.2011.10.001

[32] D. Australia, *Managing Climate and Environment Dairy Sector Food Waste Action Plan*, Dairy Australia, Southbank **2023**.

[33] S. Ghosh Ray, M. M. Ghangrekar, *Int. J. Environ. Sci. Technol.* **2019**, *16* (*1*), 527–546. DOI: https://doi.org/10.1007/s13762-018-1786-8

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[34] S. Pilli, K. Bella, Y. Manojkumar, V. Vinila, V. R. Polisetty, in *Renewable Energy Technologies for Energy Efficient Sustainable Development* (Eds: A. Sinharoy, P. N. L. Lens), Springer International Publishing, Cham **2022**.
[35] R. Girault, G. Bridoux, F. Nauleau, C. Poullain, J. Buffet, P. Peu, A. G. Sadowski, F. Béline, *Bioresour. Technol.* **2012**, *105*, 1–8. DOI: https://doi.org/10.1016/J.BIORTECH.2011.11.024
[36] M. Logan, H. Ravishankar, L. C. Tan, J. Lawrence, D. Fitzgerald, P. N. L. Lens, *Environ. Technol. Innov.* **2021**, *21*, 101352. DOI: https://doi.org/10.1016/J.ETI.2020.101352
[37] J. D. Browne, E. Allen, J. D. Murphy, *Environ. Technol.* **2013**, *34* (*13–14*), 2027–2038. DOI: https://doi.org/10.1080/09593330.2013.812669
[38] *Hill Laboratories. Certificate of Analysis No:3057205*, Hill Laboratories, Hamilton, NZ **2022**.
[39] M. Rasapoor, B. Young, R. Brar, A. Sarmah, W. Q. Zhuang, S. Baroutian, *Fuel* **2020**, *261*, 116497. DOI: https://doi.org/10.1016/J.FUEL.2019.116497
[40] Food and Agriculture Organization of the United Nations (FAO), *Law of the People's Republic of China on Prevention and Control of Water Pollution*, FAO, Rome n.d.
[41] BECA, *Biogas and Biomethane in New Zealand: Unlocking New Zealand's Renewable Natural Gas Potentia*, BECA, Duluth **2021**.
[42] Environmental Protection Agency, *Dairy Products Processing Effluent Guidelines*, Environmental Protection Agency, Washington n.d., https://www.epa.gov/eg/dairy-products-processing-effluent-guidelines
[43] European Parliament and of the Council, *Directive 2000/60/EC of the European Parliament and of the Council of 23 October 2000 Establishing a Framework for Community Action in the Field of Water Policy*, European Parliament and of the Council, Strasbourg n.d.
[44] P. Bhawan, A. Nagar, *Guidelines for Environmental Management of Dairy Farms and Gaushalas*, Central Pollution Control Board (Ministry of Environment, Forest and Climate Change, Govt. of India), New Delhi, n.d.
[45] P. Yapıcıoğlu, M. I. Yeşilnacar, *Environ. Monit. Assess.* **2020**, *192* (*8*), 536. DOI: https://doi.org/10.1007/s10661-020-08492-y
[46] J. K. Edzwald, *Water Res.* **2010**, *44* (*7*), 2077–2106. DOI: https://doi.org/10.1016/j.watres.2009.12.040
[47] H. Ødegaard, *Water Sci. Technol.* **2001**, *43* (*8*), 75–81.
[48] J. Mata-Alvarez, J. Dosta, M. S. Romero-Güiza, X. Fonoll, M. Peces, S. Astals, *Renew Sustain. Energy Rev.* **2014**, *36*, 412–427. DOI: https://doi.org/10.1016/j.rser.2014.04.039
[49] S. N. B. A. Khadaroo, P. E. Poh, D. Gouwanda, P. Grassia, *J. Environ. Chem. Eng.* **2019**, *7* (5), 103310. DOI: https://doi.org/10.1016/j.jece.2019.103310
[50] K. Rajendran, D. Mahapatra, A. V. Venkatraman, S. Muthuswamy, A. Pugazhendhi, *Renew Sustain. Energy Rev.* **2020**, *123*, 109746. DOI: https://doi.org/10.1016/j.rser.2020.109746
[51] D. G. Cirne, X. Paloumet, L. Björnsson, M. M. Alves, B. Mattiasson, *Renew Energy* **2007**, *32* (*6*), 965–975. DOI: https://doi.org/10.1016/j.renene.2006.04.003
[52] A. J. Cavaleiro, M. A. Picavet, D. Z. Sousa, A. J. M. Stams, M. A. Pereira, M. M. Alves, in *Hydrocarbon and Lipid Microbiology Protocols: Pollution Mitigation and Waste Treatment Applications* (Eds: T. J. McGenity, K. N. Timmis, B. Nogales), Springer, Berlin, Heidelberg **2017**.
[53] N. Rasit, A. Idris, R. Harun, W. A. Wan Ab Karim Ghani, *Renew Sustain. Energy Rev.* **2015**, *45*, 351–358. DOI: https://doi.org/10.1016/J.RSER.2015.01.066
[54] C. Rashama, G. Ijoma, T. Matambo, *Biomass Convers. Biorefin.* **2019**, *9* (*4*), 803–826. DOI: https://doi.org/10.1007/s13399-019-00385-6
[55] M. Carballa, W. Vestraete, in *Handbook of Hydrocarbon and Lipid Microbiology* (Ed: K. N. Timmis), Springer, Berlin, Heidelberg **2010**.
[56] M. M. Alves, M. A. Pereira, D. Z. Sousa, A. J. Cavaleiro, M. Picavet, H. Smidt, A. J. M. Stams, *Microb. Biotechnol.* **2009**, *2* (*5*), 538–550. DOI: https://doi.org/10.1111/j.1751-7915.2009.00100.x
[57] Y. Chen, J. J. Cheng, K. S. Creamer, *Bioresour. Technol.* **2008**, *99* (*10*), 4044–4064. DOI: https://doi.org/10.1016/J.BIORTECH.2007.01.057
[58] K. Hanaki, T. Matsuo, M. Nagase, *Biotechnol. Bioeng.* **1981**, *23* (*7*), 1591–1610. DOI: https://doi.org/10.1002/bit.260230717
[59] P. M. Thanh, B. Ketheesan, Z. Yan, D. Stuckey, *Biotechnol. Adv.* **2016**, *34* (*2*), 122–136. DOI: https://doi.org/10.1016/j.biotechadv.2015.12.006
[60] M. Elsamadony, A. Mostafa, M. Fujii, A. Tawfik, D. Pant, *Water Res.* **2021**, *190*, 116732. DOI: https://doi.org/10.1016/J.WATRES.2020.116732
[61] P. Sharma, A. A. Khardenavis, H. J. Purohit, in *Microbial Factories: Biofuels, Waste Treatment*, (Ed: V. C. Kalia), Springer, New Delhi **2015**.
[62] M. Ye, Y. Y. Li, *Sci. Total Environ.* **2023**, *866*, 161447. DOI: https://doi.org/10.1016/j.scitotenv.2023.161447
[63] B. C. Holohan, M. S. Duarte, M. A. Szabo-Corbacho, A. J. Cavaleiro, A. F. Salvador, M. A. Pereira, R. M. Ziels, C. T. M. J. Frijters, S. Pacheco-Ruiz, M. Carballa, D. Z. Sousa, A. J. M. Stams, V. O'Flaherty, J. B. Van Lier, M. M. Alves, *Environ. Sci. Technol.* **2022**, *56* (*8*), 4749–4775. DOI: https://doi.org/10.1021/acs.est.1c08722
[64] E. J. Martínez, V. Redondas, J. Fierro, X. Gómez, A. Morán, *J. Residuals Sci. Technol.* **2011**, *8* (*2*), 53–60.
[65] R. T. Haug, D. C. Stuckey, J. M. Gossett, P. L. McCarty, *J. Water Pollut. Control Fed.* **1978**, *50* (*1*), 73–85.
[66] F. M. Müller, D. de Oliveira, C. Michels, *J. Water Process Eng.* **2023**, *52*, 103480. DOI: https://doi.org/10.1016/j.jwpe.2022.103480
[67] A. E. Berninghaus, T. S. Radniecki, *Environ. Eng. Sci.* **2022**, *39* (*6*), 512–522. DOI: https://doi.org/10.1089/ees.2021.0119
[68] H. Tian, P. Karachalios, I. Angelidaki, I. A. Fotidis, *Chem. Eng. J.* **2018**, *349*, 574–580. DOI: https://doi.org/10.1016/J.CEJ.2018.05.083
[69] M. Ye, A. Zhu, B. Sun, Y. Qin, Y.-Y. Li, *Bioresour. Technol.* **2022**, *357*, 127349. DOI: https://doi.org/10.1016/j.biortech.2022.127349
[70] S. Saha, B.-H. Jeon, M. B. Kurade, P. K. Chatterjee, S. W. Chang, K. Markkandan, E.-S. Salama, S. P. Govindwar, H.-S. Roh, *Chem. Eng. J.* **2019**, *358*, 188–196. DOI: https://doi.org/10.1016/j.cej.2018.09.220
[71] Y. Y. Choong, I. Norli, A. Z. Abdullah, M. F. Yhaya, *Bioresour. Technol.* **2016**, *209*, 369–379. DOI: https://doi.org/10.1016/j.biortech.2016.03.028
[72] N. Ganidi, S. Tyrrel, E. Cartmell, *Bioresour. Technol.* **2009**, *100* (*23*), 5546–5554. DOI: https://doi.org/10.1016/j.biortech.2009.06.024

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[73] A. Noyola, J. M. Morgan-Sagastume, J. E. Lopez-Hernandez, *Rev. Environ. Sci. Biotechnol.* **2006**, *5* (*1*), 93–114. DOI: https://doi.org/10.1007/s11157-005-2754-6.

[74] M. S. Islam, V. V Ranade, *Chem. Eng. Process. Process Intensif.* **2024**, *198*, 109733. DOI: https://doi.org/10.1016/j.cep.2024.109733

[75] E. Leca, B. Zennaro, J. Hamelin, H. Carrère, C. Sambusiti, *Biotechnol. Adv.* **2023**, *65*, 108129. DOI: https://doi.org/10.1016/j.biotechadv.2023.108129

[76] N. Thakur, M. Jalalah, S. A. Alsareii, F. A. Harraz, A. A. Almadiy, S. Su, E.-S. Salama, X. Li, *Renew Sustain. Energy Rev.* **2024**, *191*, 114155. DOI: https://doi.org/10.1016/j.rser.2023.114155

[77] R. M. Alqaralleh, K. Kennedy, R. Delatolla, *J. Environ. Manage.* **2018**, *217*, 416–428. DOI: https://doi.org/10.1016/j.jenvman.2018.03.123

[78] S. Luste, S. Luostarinen, M. Sillanpää, *J. Hazard. Mater.* **2009**, *164* (*1*), 247–255. DOI: https://doi.org/10.1016/J.JHAZMAT.2008.08.002

[79] P. W. Harris, T. Schmidt, B. K. McCabe, *Bioresour. Technol.* **2017**, *244*, 605–610. DOI: https://doi.org/10.1016/j.biortech.2017.07.179

[80] H. Carrère, C. Dumas, A. Battimelli, D. J. Batstone, J. P. Delgenès, J. P. Steyer, I. Ferrer, *J. Hazard. Mater.* **2010**, *183* (*1–3*), 1–15. DOI: https://doi.org/10.1016/J.JHAZMAT.2010.06.129

[81] Z. Liang, N. Shen, C. Lu, Y. Chen, Y. Guan, *Biomass Bioenergy* **2021**, *151*, 106176. DOI: https://doi.org/10.1016/j.biombioe.2021.106176

[82] W. Ren, Y. Zhang, X. Liu, S. Li, H. Li, Y. Zhai, *J. Environ. Manage.* **2024**, *349*, 119427. DOI: https://doi.org/10.1016/j.jenvman.2023.119427

[83] D. C. Devlin, S. R. R. Esteves, R. M. Dinsdale, A. J. Guwy, *Bioresour. Technol.* **2011**, *102* (*5*), 4076–4082. DOI: https://doi.org/10.1016/j.biortech.2010.12.043

[84] W. Wei, Q. Wang, L. Zhang, A. Laloo, H. Duan, D. J. Batstone, Z. Yuan, *Water Res.* **2018**, *130*, 13–19. DOI: https://doi.org/10.1016/j.watres.2017.11.050

[85] Z. Liang, N. Shen, C. Lu, Y. Chen, Y. Guan, *Biomass Bioenergy* **2021**, *151*, 106176. DOI: https://doi.org/10.1016/j.biombioe.2021.106176

[86] A. G. Vlyssides, P. K. Karlis, *Bioresour. Technol.* **2004**, *91* (*2*), 201–206. DOI: https://doi.org/10.1016/S0960-8524(03)00176-7

[87] J. Kim, C. Park, T.-H. Kim, M. Lee, S. Kim, S.-W. Kim, J. Lee, *J. Biosci. Bioeng.* **2003**, *95* (*3*), 271–275. DOI: https://doi.org/10.1016/S1389-1723(03)80028-2

[88] A. Valo, H. Carrère, J. P. Delgenès, *J. Chem. Technol. Biotechnol.* **2004**, *79* (*11*), 1197–1203. DOI: https://doi.org/10.1002/jctb.1106

[89] Gh. Hassan Dar, S. M. Tandon, *Biol. Wastes* **1987**, *21* (*2*), 75–83. DOI: https://doi.org/10.1016/0269-7483(87)90131-5

[90] S. Dong, S. Asli, K. Hur, N. Massalha, *Bioresour. Technol. Rep.* **2024**, *27*, 101879. DOI: https://doi.org/10.1016/j.biteb.2024.101879

[91] M. Weemaes, H. Grootaerd, F. Simoens, W. Verstraete, *Water Res.* **2000**, *34* (*8*), 2330–2336. DOI: https://doi.org/10.1016/S0043-1354(99)00373-5

[92] M. Hodaei, S. Ghasemi, A. Khosravi, M. Vossoughi, *Biomass Bioenergy* **2021**, *152*, 106198. DOI: https://doi.org/10.1016/j.biombioe.2021.106198

[93] C. Bougrier, C. Albasi, J. P. Delgenès, H. Carrère, *Chem. Eng. Process. Process Intensif.* **2006**, *45* (*8*), 711–718. DOI: https://doi.org/10.1016/j.cep.2006.02.005

[94] J. W. Lim, J.-Y. Wang, *Waste Manage.* **2013**, *33* (*4*), 813–819. DOI: https://doi.org/10.1016/j.wasman.2012.11.013

[95] S. Montalvo, C. Huiliñir, F. Ojeda, A. Castillo, L. Lillo, L. Guerrero, *Int. Biodeterior. Biodegrad.* **2016**, *110*, 1–7. DOI: https://doi.org/10.1016/j.ibiod.2016.01.010

[96] S. Hasegawa, N. Shiota, K. Katsura, A. Akashi, *Water Sci. Technol.* **2000**, *41* (*3*), 163–169. DOI: https://doi.org/10.2166/wst.2000.0068

[97] Z. Fan, in *Proceedings of 2022 7th International Conference on Environmental Engineering and Sustainable Development (CEESD 2022)* (Ed: G. Huang), Springer International Publishing, Cham **2023**.

[98] T. S. O. Souza, A. Carvajal, A. Donoso-Bravo, M. Peña, F. Fdz-Polanco, *Water Res.* **2013**, *47* (*9*), 3244–3254. DOI: https://doi.org/10.1016/j.watres.2013.03.041

[99] M. Barjenbruch, O. Kopplow, *Adv. Environ. Res.* **2003**, *7* (*3*), 715–720. DOI: https://doi.org/10.1016/S1093-0191(02)00032-1

[100] N. C. Pascale, J. J. Chastinet, D. M. Bila, G. L. SantAnna jr., S. L. Quitério, S. M. R. Vendramel, *Water Sci. Technol.* **2018**, *79* (*5*), 985–992. DOI: https://doi.org/10.2166/wst.2018.508

[101] Y. Yin, Y.-J. Liu, S.-J. Meng, E. U. Kiran, Y. Liu, *Appl. Energy* **2016**, *179*, 1131–1137. DOI: https://doi.org/10.1016/j.apenergy.2016.07.083

[102] A. A. Mendes, E. B. Pereira, H. F. de Castro, *Biochem. Eng. J.* **2006**, *32* (*3*), 185–190. DOI: https://doi.org/10.1016/j.bej.2006.09.021

[103] Å. Davidsson, J. Wawrzynczyk, O. Norrlöw, J. L. C. Jansen, *J. Residuals Sci. Technol.* **2007**, *4* (*1*), 1–7.

[104] X. Liu, Q. Wang, Y. Tang, S. G. Pavlostathis, *Chem. Eng. J.* **2021**, *410*, 127430. DOI: https://doi.org/10.1016/j.cej.2020.127430

[105] D. Kim, K. Lee, K. Y. Park, *Int. Biodeterior. Biodegrad.* **2015**, *101*, 42–46. DOI: https://doi.org/10.1016/j.ibiod.2015.03.025

[106] M. Park, N. Kim, S. Jung, T.-Y. Jeong, D. Park, *Chemosphere* **2021**, *264*, 128516. DOI: https://doi.org/10.1016/j.chemosphere.2020.128516

[107] J.-M. Choi, S.-K. Han, C.-Y. Lee, *Bioresour. Technol.* **2018**, *259*, 207–213. DOI: https://doi.org/10.1016/j.biortech.2018.02.123

[108] T. M. Thompson, B. R. Young, S. Baroutian, *Chemosphere* **2021**, *275*, 130035. DOI: https://doi.org/10.1016/j.chemosphere.2021.130035

[109] P. L. Ngo, B. R. Young, K. Brian, S. Baroutian, *J. Clean. Prod.* **2023**, *428*, 139354. DOI: https://doi.org/10.1016/j.jclepro.2023.139354

[110] K. Wu, W. Xu, J. Lu, C. Wang, J. Liao, X. He, *Renew Energy* **2022**, *195*, 1103–1112. DOI: https://doi.org/10.1016/j.renene.2022.06.055

[111] A. H. Mouneimne, H. Carrère, N. Bernet, J. P. Delgenès, *Bioresour. Technol.* **2003**, *90* (*1*), 89–94. DOI: https://doi.org/10.1016/S0960-8524(03)00091-9

[112] S. Tanaka, T. Kobayashi, K. Kamiyama, M. L. N. Signey Bildan, *Water Sci. Technol.* **1997**, *35* (*8*), 209–215. DOI: https://doi.org/10.1016/S0273-1223(97)00169-8

[113] J. Liu, M. Yang, J. Zhang, J. Zheng, H. Xu, Y. Wang, Y. Wei, *Bioresour. Technol.* **2018**, *256*, 118–127. DOI: https://doi.org/10.1016/j.biortech.2018.01.126

[114] B. T. Ray, J.-G. Lin, R. V Rajan, *Res. J. Water Pollut. Control Fed.* **1990**, *62* (*1*), 81–87.

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[115] C. Park, C. Lee, S. Kim, Y. Chen, H. A. Chase, *J. Biosci. Bioeng.* **2005**, *100* (*2*), 164–167. DOI: https://doi.org/10.1263/jbb.100.164

[116] S. Pérez-Elvira, M. Fdz-Polanco, F. I. Plaza, G. Garralón, F. Fdz-Polanco, *Water Sci. Technol.* **2009**, *60* (*6*), 1525–1532. DOI: https://doi.org/10.2166/wst.2009.484

[117] W. P. Barber, *Water Environ. J.* **2005**, *19* (*1*), 2–7. DOI: https://doi.org/10.1111/j.1747-6593.2005.tb00542.x

[118] U. Neis, K. Nickel, A. Lundén, *J. Environ. Sci. Health. Part A* **2008**, *43* (*13*), 1541–1545. DOI: https://doi.org/10.1080/10934520802293701

[119] C. Bougrier, H. Carrère, J. P. Delgenès, *Chem. Eng. J.* **2005**, *106* (*2*), 163–169. DOI: https://doi.org/10.1016/j.cej.2004.11.013

[120] P. Roebuck, K. Kennedy, R. Delatolla, *Water Qual. Res. J.* **2019**, *54* (*4*), 265–277. DOI: https://doi.org/10.2166/wqrj.2019.039

[121] P. Neumann, Z. González, G. Vidal, *Bioresour. Technol.* **2017**, *234*, 178–187. DOI: https://doi.org/10.1016/j.biortech.2017.03.029

[122] T. I. Onyeche, O. Schläfer, H. Bormann, C. Schröder, M. Sievers, *Ultrasonics* **2002**, *40* (*1*), 31–35. DOI: https://doi.org/10.1016/S0041-624X(02)00087-2

[123] S. Mirmasoumi, S. Ebrahimi, R. K. Saray, *Energy* **2018**, *157*, 707–717. DOI: https://doi.org/10.1016/j.energy.2018.06.003

[124] A. C. Lizama, C. C. Figueiras, R. R. Herrera, A. Z. Pedreguera, J. E. Ruiz Espinoza, *Int. Biodeterior. Biodegrad.* **2017**, *123*, 1–9. DOI: https://doi.org/10.1016/j.ibiod.2017.05.020

[125] M. R. Salsabil, A. Prorot, M. Casellas, C. Dagot, *Chem. Eng. J.* **2009**, *148* (*2*), 327–335. DOI: https://doi.org/10.1016/j.cej.2008.09.003

[126] M. Zieliński, M. Dębowski, M. Kisielewska, A. Nowicka, M. Rokicka, K. Szwarc, *Waste Biomass Valorization* **2019**, *10* (*4*), 747–754. DOI: https://doi.org/10.1007/s12649-017-9977-y

[127] C. Hubert, B. Steiniger, C. Schaum, *Water Environ. Res.* **2020**, *92* (*4*), 534–540. DOI: https://doi.org/10.1002/wer.1197

[128] L. Ferreira, E. Duarte, D. Figueiredo, *Bioresour. Technol.* **2012**, *116*, 285–289. DOI: https://doi.org/10.1016/j.biortech.2012.03.063

[129] L. Wang, T. N. Aziz, F. L. De Los Reyes, *Water Res.* **2013**, *47* (*11*), 3835–3844. DOI: https://doi.org/10.1016/j.watres.2013.04.003

[130] C. Wan, Q. Zhou, G. Fu, Y. Li, *Waste Manage.* **2011**, *31* (*8*), 1752–1758. DOI: https://doi.org/10.1016/j.wasman.2011.03.025

[131] G. Silvestre, A. Rodríguez-Abalde, B. Fernández, X. Flotats, A. Bonmatí, *Bioresour. Technol.* **2011**, *102* (*13*), 6830–6836. DOI: https://doi.org/10.1016/j.biortech.2011.04.019

[132] M. Philipp, H. Ackermann, N. Barbana, J. Pluschke, S. U. Geißen, *Water (Basel)* **2023**, *15* (*10*), 1818. DOI: https://doi.org/10.3390/w15101818

[133] J. Gelegenis, D. Georgakakis, I. Angelidaki, N. Christopoulou, M. Goumenaki, *Appl. Energy* **2007**, *84* (*6*), 646–663. DOI: https://doi.org/10.1016/j.apenergy.2006.12.001

[134] S. Borowski, P. Kubacki, *Waste Manage.* **2015**, *40*, 119–126. DOI: https://doi.org/10.1016/j.wasman.2015.03.021

[135] S. Luostarinen, S. Luste, M. Sillanpää, *Bioresour. Technol.* **2009**, *100* (*1*), 79–85. DOI: https://doi.org/10.1016/j.biortech.2008.06.029

[136] Å. Davidsson, C. Lövstedt, J. la Cour Jansen, C. Gruvberger, H. Aspegren, *Waste Manage.* **2008**, *28* (*6*), 986–992. DOI: https://doi.org/10.1016/J.WASMAN.2007.03.024

[137] V. Razaviarani, I. D. Buchanan, S. Malik, H. Katalambula, *J. Environ. Manage.* **2013**, *123*, 26–33. DOI: https://doi.org/10.1016/j.jenvman.2013.03.021

[138] M. S. Romero-Güiza, J. Vila, J. Mata-Alvarez, J. M. Chimenos, S. Astals, *Renew Sustain. Energy Rev.* **2016**, *58*, 1486–1499. DOI: https://doi.org/10.1016/J.RSER.2015.12.094

[139] M. Ye, J. Liu, C. Ma, Y.-Y. Li, L. Zou, G. Qian, Z. P. Xu, *J. Clean. Prod.* **2018**, *192*, 316–326. DOI: https://doi.org/10.1016/j.jclepro.2018.04.244

[140] J. Zhang, R. Zhang, H. Wang, K. Yang, *Sci. Total Environ.* **2020**, *704*, 135282. DOI: https://doi.org/10.1016/j.scitotenv.2019.135282

[141] A. H. Mouneimne, H. Carrère, N. Bernet, J. P. Delgenès, *Environ. Technol.* **2004**, *25* (*4*), 459–469. DOI: https://doi.org/10.1080/09593332508618455

[142] H. Zhu, Y. Han, W. Ma, H. Han, W. Ma, C. Xu, *Bioresour. Technol.* **2018**, *262*, 302–309. DOI: https://doi.org/10.1016/j.biortech.2018.04.080

[143] C. Zhang, S. Yun, X. Li, Z. Wang, H. Xu, T. Du, *Bioresour. Technol.* **2018**, *263*, 517–524. DOI: https://doi.org/10.1016/j.biortech.2018.05.042

[144] L. Zhang, K.-C. Loh, *Bioresour. Technol.* **2019**, *278*, 108–115. DOI: https://doi.org/10.1016/j.biortech.2019.01.073

[145] J. Palatsi, R. Affes, B. Fernandez, M. A. Pereira, M. M. Alves, X. Flotats, *Water Res.* **2012**, *46* (*16*), 5268–5278. DOI: https://doi.org/10.1016/j.watres.2012.07.008

[146] J. Ma, M. Amjad Bashir, J. Pan, L. Qiu, H. Liu, L. Zhai, A. Rehim, *J. Clean. Prod.* **2018**, *183*, 11–19. DOI: https://doi.org/10.1016/j.jclepro.2018.02.121

[147] F. Hu, S. Zhang, X. Wang, C. Wang, J. Wu, L. Xu, G. Xu, Y. Hu, *Biochem. Eng. J.* **2022**, *184*, 108490. DOI: https://doi.org/10.1016/j.bej.2022.108490

[148] M. O. Fagbohungbe, B. M. J. Herbert, L. Hurst, H. Li, S. Q. Usmani, K. T. Semple, *Bioresour. Technol.* **2016**, *216*, 142–149. DOI: https://doi.org/10.1016/j.biortech.2016.04.106

[149] Z. Zhao, Y. Zhang, T. L. Woodard, K. P. Nevin, D. R. Lovley, *Bioresour. Technol.* **2015**, *191*, 140–145. DOI: https://doi.org/10.1016/j.biortech.2015.05.007

[150] C. Luo, F. Lü, L. Shao, P. He, *Water Res.* **2015**, *68*, 710–718. DOI: https://doi.org/10.1016/j.watres.2014.10.052

[151] H. M. Jang, Y.-K. Choi, E. Kan, *Bioresour. Technol.* **2018**, *250*, 927–931. DOI: https://doi.org/10.1016/j.biortech.2017.11.074

[152] Z. Zhao, Y. Zhang, Y. Li, Y. Dang, T. Zhu, X. Quan, *Chem. Eng. J.* **2017**, *313*, 10–18. DOI: https://doi.org/10.1016/j.cej.2016.11.149

[153] Y. Lei, D. Sun, Y. Dang, H. Chen, Z. Zhao, Y. Zhang, D. E. Holmes, *Bioresour. Technol.* **2016**, *222*, 270–276. DOI: https://doi.org/10.1016/j.biortech.2016.10.007

[154] Y. Dang, D. Sun, T. L. Woodard, L.-Y. Wang, K. P. Nevin, D. E. Holmes, *Bioresour. Technol.* **2017**, *238*, 30–38. DOI: https://doi.org/10.1016/j.biortech.2017.04.021

[155] Y. Dang, D. E. Holmes, Z. Zhao, T. L. Woodard, Y. Zhang, D. Sun, L.-Y. Wang, K. P. Nevin, D. R. Lovley, *Bioresour. Technol.* **2016**, *220*, 516–522. DOI: https://doi.org/10.1016/j.biortech.2016.08.114

[156] L.-L. Li, Z.-H. Tong, C.-Y. Fang, J. Chu, H.-Q. Yu, *Water Res.* **2015**, *70*, 1–8. DOI: https://doi.org/10.1016/j.watres.2014.11.042

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[157] E. E. Ziganshina, S. S. Bulynina, A. M. Ziganshin, *Fermentation* **2022**, *8* (*11*), 641. DOI: https://doi.org/10.3390/fermentation8110641

[158] L. Liu, S. Yun, T. Ke, K. Wang, J. An, J. Liu, *Waste Manage.* **2023**, *159*, 163–173. DOI: https://doi.org/10.1016/j.wasman.2023.01.036

[159] Y. Hao, Y. Wang, C. Ma, J. C. White, Z. Zhao, C. Duan, Y. Zhang, M. Adeel, Y. Rui, G. Li, B. Xing, *J. Clean. Prod.* **2019**, *240*, 118257. DOI: https://doi.org/10.1016/j.jclepro.2019.118257

[160] S. Yang, S. Han, Y.-M. Yun, S. Kang, *Minerals* **2021**, *11* (*2*), 179. DOI: https://doi.org/10.3390/min11020179

[161] R. Lin, J. Cheng, J. Zhang, J. Zhou, K. Cen, J. D. Murphy, *Bioresour. Technol.* **2017**, *239*, 345–352. DOI: https://doi.org/10.1016/j.biortech.2017.05.017

[162] T. Tian, S. Qiao, X. Li, M. Zhang, J. Zhou, *Bioresour. Technol.* **2017**, *224*, 41–47. DOI: https://doi.org/10.1016/j.biortech.2016.10.058

[163] P. Wang, Y. Zheng, P. Lin, J. Li, H. Dong, H. Yu, L. Qi, L. Ren, *Bioresour. Technol.* **2021**, *339*, 125585. DOI: https://doi.org/10.1016/j.biortech.2021.125585

[164] M. Basitere, M. Njoya, S. K. O. Ntwampe, M. S. Sheldon, *Water Pract. Technol.* **2020**, *15* (*2*), 248–260. DOI: https://doi.org/10.2166/wpt.2020.023

[165] J. Ryue, L. Lin, F. L. Kakar, E. Elbeshbishy, A. Al-Mamun, B. R. Dhar, *Energy Sustainable Dev.* **2020**, *54*, 72–84. DOI: https://doi.org/10.1016/j.esd.2019.11.001

[166] J. C. Kabouris, U. Tezel, S. G. Pavlostathis, M. Engelmann, A. C. Todd, R. A. Gillette, *Water Environ. Res.* **2008**, *80* (*3*), 212–221. DOI: https://doi.org/10.2175/106143007×220699

[167] C. Li, P. Champagne, B. C. Anderson, *Environ. Technol.* **2013**, *34* (*13–14*), 2125–2133. DOI: https://doi.org/10.1080/09593330.2013.824010

[168] X. Guo, C. Wang, F. Sun, W. Zhu, W. Wu, *Bioresour. Technol.* **2014**, *152*, 420–428. DOI: https://doi.org/10.1016/j.biortech.2013.11.012

[169] M. Kim, Y.-H. Ahn, R. E. Speece, *Water Res.* **2002**, *36* (*17*), 4369–4385. DOI: https://doi.org/10.1016/S0043-1354(02)00147-1

[170] Y. Meng, C. Jost, J. Mumme, K. Wang, B. Linke, *Chem. Eng. J.* **2016**, *288*, 79–86. DOI: https://doi.org/10.1016/j.cej.2015.11.072

[171] M. Pohl, J. Mumme, K. Heeg, E. Nettmann, *Bioresour. Technol.* **2012**, *124*, 321–327. DOI: https://doi.org/10.1016/j.biortech.2012.08.063

[172] W. M. Budzianowski, *Renew Sustain. Energy Rev.* **2016**, *54*, 1148–1171. DOI: https://doi.org/10.1016/J.RSER.2015.10.054

[173] K. Sudalyandi, R. Jeyakumar, in *Biofuel Production Using Anaerobic Digestion*, Springer, Singapore **2022**.

[174] G. Srisowmeya, M. Chakravarthy, G. Nandhini Devi, *Renew Sustain. Energy Rev.* **2020**, *119*, 109587. DOI: https://doi.org/10.1016/j.rser.2019.109587

[175] B. Chatterjee, D. Mazumder, *Renew Sustain. Energy Rev.* **2019**, *104*, 439–469. DOI: https://doi.org/10.1016/j.rser.2019.01.026

[176] C. Akobi, H. Yeo, H. Hafez, G. Nakhla, *Appl. Energy* **2016**, *184*, 548–559. DOI: https://doi.org/10.1016/j.apenergy.2016.10.039

[177] E. Y. Lim, H. Tian, Y. Chen, K. Ni, J. Zhang, Y. W. Tong, *Bioresour. Technol.* **2020**, *314*, 123751. DOI: https://doi.org/10.1016/j.biortech.2020.123751

[178] L. Li, S. Yuan, C. Cai, X. Dai, *Front. Environ. Sci. Eng.* **2021**, *16* (*6*), 74. DOI: https://doi.org/10.1007/s11783-021-1508-1

[179] Y. Li, M. Gao, D. Hua, J. Zhang, Y. Zhao, H. Mu, H. Xu, X. Liang, F. Jin, X. Zhang, *Ann. Microbiol.* **2015**, *65* (*3*), 1465–1471. DOI: https://doi.org/10.1007/s13213-014-0985-x

[180] L.-J. Wu, T. Kobayashi, Y.-Y. Li, K.-Q. Xu, *Energy Convers. Manag.* **2015**, *106*, 1174–1182. DOI: https://doi.org/10.1016/j.enconman.2015.10.059

[181] E. Holl, J. Steinbrenner, W. Merkle, J. Krümpel, S. Lansing, U. Baier, H. Oechsner, A. Lemmer, *Bioresour. Technol.* **2022**, *360*, 127633. DOI: https://doi.org/10.1016/j.biortech.2022.127633

[182] F. Micolucci, M. Gottardo, P. Pavan, C. Cavinato, D. Bolzonella, *J. Clean. Prod.* **2018**, *171*, 1376–1385. DOI: https://doi.org/10.1016/j.jclepro.2017.10.080

[183] Y. Jo, J. Kim, K. Hwang, C. Lee, *Waste Manage.* **2018**, *78*, 509–520. DOI: https://doi.org/10.1016/j.wasman.2018.06.017

[184] FAO, *Dairy Market Review: Emerging Trends and Outlook 2022*, FAO, Rome **2022**.

[185] K. Miettinen, F. Ruiz, A. P. Wierzbicki, in *Multiobjective Optimization: Interactive and Evolutionary Approaches* (Eds: J. Branke, K. Deb, K. Miettinen, R. Słowiński), Springer, Berlin, Heidelberg **2008**.

[186] J. R. S. C. Mateo, in *Multi Criteria Analysis in the Renewable Energy Industry* (Ed: J. R. San Cristóbal Mateo), Springer, London **2012**.

[187] A. Jadhav, R. Sonar, in *2009 Second Int Conf Emerging Trends Eng Technol*, IEEE, New York **2009**.

[188] Q. Chen, W. Wu, D. Qi, Y. Ding, Z. Zhao, *Sci. Total Environ.* **2020**, *710*, 136388. DOI: https://doi.org/10.1016/j.scitotenv.2019.136388

[189] D. Nguyen, S. K. Khanal, *Biotechnol. Adv.* **2018**, *36* (*7*), 1971–1983. DOI: https://doi.org/10.1016/j.biotechadv.2018.08.007

[190] R. Datta, J. Holatko, O. Latal, T. Hammerschmiedt, J. Elbl, V. Pecina, A. Kintl, L. Balakova, M. Radziemska, T. Baltazar, P. Skarpa, S. Danish, M. Zafar-ul-Hye, T. Vyhnanek, M. Brtnicky, *Land (Basel)* **2020**, *9* (*8*): 258. DOI: https://doi.org/10.3390/land9080258

[191] B. Klik, J. Holatko, I. Jaskulska, M. Z. Gusiatin, T. Hammerschmiedt, M. Brtnicky, E. Liniauskienė, T. Baltazar, D. Jaskulski, A. Kintl, M. Radziemska, *Materials* **2022**, *15* (*23*) 8331. DOI: https://doi.org/10.3390/ma15238331

[192] J. Mi, E. G. Gregorich, S. Xu, N. B. McLaughlin, J. Liu, *Sci. Rep.* **2020**, *10* (*1*), 18282. DOI: https://doi.org/10.1038/s41598-020-75350-9

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